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DATA SCIENCE AND ARTIFICIAL INTELLIGENCE

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# TITOLO

# Inflation Forecasting with Macroeconomic Indicators: Assessing the Contribution of News Sentiment

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# Abstract

This thesis, developed in collaboration with Generali Insurance Company, assesses whether news-based indicators add predictive information beyond inflation lags and macroeconomic variables for month-on-month inflation in the European Union's fixed 27-country aggregate (EU-27). The project also includes a prototype analytical dashboard intended to support the company's economic analysis.

The European analysis combines Harmonised Index of Consumer Prices (HICP) data, macroeconomic indicators and selected Guardian headlines and short summaries over 2007–2025. FinBERT, a language model adapted to financial sentiment analysis, is used to construct monthly sentiment measures and topic-specific negative-news indicators covering energy, supply chains and geopolitical developments. News published during a given month is combined with lagged inflation and macroeconomic observations to estimate that month's inflation at month-end, ahead of the final HICP release in the following month. This retrospective exercise uses historical data without reconstructing release dates or data vintages.

An autoregressive hidden Markov model (AR-HMM) is compared with linear autoregressive (AR), Random Forest and gradient boosting benchmarks under matched information sets. European model selection uses validation over 2017–2021, with an initial single-block structure comparison and later validation blocks with expanding training samples. The final test covers 2022–2025. A supplementary analysis using U.S. Consumer Price Index (CPI) data and New York Times news provides exploratory context and has no separate validation stage.

In the matched European final test, the AR-HMM combining lagged inflation, macroeconomic predictors and news achieves the lowest root mean squared error (RMSE) among the 12 specifications, at 0.3959 percentage points. Adding news reduces RMSE by approximately 11.2% relative to the macroeconomic-only AR-HMM (0.4460). This reduction is statistically significant (Holm-adjusted $p = 0.018$). The results demonstrate the incremental predictive value of target-month news for this specification over 2022–2025 and support further assessment through the company's analytical dashboard.

# Abstract in italiano

Questa tesi, sviluppata in collaborazione con Generali Insurance Company, valuta se gli indicatori ricavati dalle notizie apportino informazione predittiva aggiuntiva rispetto ai valori ritardati dell'inflazione e alle variabili macroeconomiche per l'inflazione su base mensile nell'aggregato a composizione fissa dei 27 Stati membri dell'Unione europea (UE-27). Il progetto comprende inoltre un prototipo di dashboard analitica destinato a supportare l'analisi economica dell'azienda.

L'analisi europea combina dati dell'Indice armonizzato dei prezzi al consumo (HICP), indicatori macroeconomici e titoli e brevi sommari selezionati di The Guardian nel periodo 2007–2025. FinBERT, un modello linguistico adattato all'analisi del sentiment finanziario, viene utilizzato per costruire misure mensili di sentiment e indicatori di notizie negative relativi a energia, catene di approvvigionamento e sviluppi geopolitici. Le notizie pubblicate in un determinato mese vengono combinate con osservazioni ritardate dell'inflazione e delle variabili macroeconomiche per stimare, a fine mese, l'inflazione dello stesso mese, prima della pubblicazione dei dati HICP definitivi nel mese successivo. Questo esercizio retrospettivo utilizza dati storici senza ricostruire le date di pubblicazione o le versioni dei dati disponibili alle diverse date del passato.

Un modello di Markov nascosto autoregressivo (AR-HMM) viene confrontato con modelli autoregressivi lineari (AR), Random Forest e gradient boosting, utilizzando gli stessi insiemi informativi. La selezione dei modelli europei si basa sulla validazione nel periodo 2017–2021, con un confronto iniziale delle strutture su un unico blocco e successivi blocchi di validazione con campioni di addestramento crescenti. Il test finale copre il periodo 2022–2025. Un'analisi integrativa basata sui dati dell'Indice dei prezzi al consumo (CPI) degli Stati Uniti e sulle notizie di The New York Times fornisce un confronto esplorativo e non prevede una fase di validazione separata.

Nel test finale europeo, l'AR-HMM che combina valori ritardati dell'inflazione, indicatori macroeconomici e notizie ottiene la più bassa radice dell'errore quadratico medio (RMSE) tra le 12 specificazioni, pari a 0,3959 punti percentuali. L'aggiunta delle notizie riduce l'RMSE di circa l'11,2% rispetto all'AR-HMM con i soli predittori macroeconomici aggiuntivi (0,4460). La riduzione è statisticamente significativa (p corretto con il metodo di Holm = 0,018). I risultati dimostrano il valore predittivo aggiuntivo delle notizie del mese corrente per questa specificazione nel periodo 2022–2025 e ne sostengono l'ulteriore valutazione attraverso la dashboard analitica dell'azienda.

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# List of Abbreviations

AIC - Akaike information criterion
API - Application programming interface
AR - Autoregressive
AR-HMM - Autoregressive hidden Markov model
ARX - Autoregressive model with exogenous predictors
BERT - Bidirectional Encoder Representations from Transformers
BIC - Bayesian information criterion
CPI - Consumer Price Index
CPU - Central processing unit
CSV - Comma-separated values
ECB - European Central Bank
EU-27 - European Union aggregate of 27 member states
FRED - Federal Reserve Economic Data
GDP - Gross domestic product
HICP - Harmonised Index of Consumer Prices
HMM - Hidden Markov model
HTML - Hypertext Markup Language
MAE - Mean absolute error
ME - Mean error
MS - Markov switching
NLP - Natural language processing
OLS - Ordinary least squares
PPI - Producer Price Index
RMSE - Root mean squared error
RMSFE - Root mean squared forecast error
SA - Seasonally adjusted
SQL - Structured Query Language
WTI - West Texas Intermediate

# Chapter 1 Introduction

## 1.1 Background and Estimation Task

Developed in collaboration with Generali Insurance Company, this thesis examines whether news indicators improve monthly inflation estimates beyond the information in past inflation and macroeconomic variables. Its main application is the European Union's fixed 27-country aggregate (EU-27). Inflation is measured as the month-on-month percentage change in the all-items Harmonised Index of Consumer Prices (HICP), which supports comparisons of consumer-price developments across European economies [1]. A prototype analytical dashboard brings the results together for company analysts working on economic questions relevant to asset allocation and portfolio management.

The thesis compares an autoregressive hidden Markov model (AR-HMM) with linear autoregressive (AR), Random Forest and gradient boosting benchmarks. The AR-HMM uses a regime-switching framework [2]. In the principal European specification, the intercept and autoregressive coefficients vary across latent states, while the innovation variance is common. Alternative switching restrictions are examined separately. The latent states describe fitted inflation dynamics. Their economic interpretation is discussed alongside the results.

The analysis tests the incremental contribution of news indicators alongside macroeconomic predictors. FinBERT, a BERT-based language model adapted to financial sentiment analysis [3], is applied to selected news-text fields to construct monthly sentiment measures. These include general sentiment and topic-specific negative-news indicators for energy, supply chains and geopolitical developments. The European corpus comes from The Guardian, while the supplementary U.S. corpus comes from The New York Times. These indicators summarise the financial tone and topic content of the selected articles.

The practical aim is to estimate inflation for a given month at month-end, ahead of the final HICP release in the following month [1]. News indicators summarise articles published during the month being estimated, while inflation and macroeconomic predictors use earlier observations at the experiment-specific lags described in Chapter 3. These lags implement the intended reporting-delay convention. Historical release dates and data vintages are not reconstructed, so the empirical exercise remains a retrospective month-end nowcast rather than a verified real-time estimate.

Previous studies report mixed evidence on the usefulness of news sentiment for inflation prediction [4, 5]. Their findings concern different countries, inflation measures, horizons and benchmarks. This motivates a setting-specific question: whether Guardian-derived news indicators add predictive information beyond inflation lags and macroeconomic predictors for EU-27 monthly inflation. The thesis addresses this question through comparisons under

matched predictor sets and evaluation periods. Accuracy is assessed primarily by root mean squared error (RMSE), which gives greater weight to large estimation errors, and by mean absolute error (MAE), which summarises their average absolute magnitude. Both metrics are expressed in percentage points of inflation.

### 1.2 Research Questions and Contributions

The European analysis centres on the contribution of news to monthly inflation estimation. Three research questions distinguish the choice of model, the treatment of inflation dynamics and the value of additional predictors:

RQ1. Under matched predictor sets and evaluation periods, does AR-HMM estimate EU-27 month-on-month inflation more accurately than linear AR, Random Forest and gradient boosting?

RQ2. How do regime structure, inflation lag order and calendar-month centring affect AR-HMM estimation accuracy?

RQ3. Do macroeconomic predictors improve accuracy beyond inflation lags, and do target-month news indicators provide further gains beyond inflation lags and macroeconomic predictors?

The project makes three contributions. First, it constructs a monthly dataset combining EU-27 HICP, macroeconomic series and dated news, with sources and processing steps documented to support reproducibility. Second, it compares model structures and predictor groups over matched European evaluation periods to assess the incremental predictive value of macroeconomic and news information. Third, it provides a prototype analytical dashboard for company use. A supplementary U.S. analysis provides exploratory international context.

### 1.3 Scope and Structure

The principal dataset covers the EU-27 aggregate over 2007–2025. European model selection uses validation over 2017–2021. Initial structure selection evaluates a single validation block; later comparisons use three successive validation blocks with expanding training samples and models fitted once per block. The final test covers January 2022–December 2025. Chapter 4 details this schedule. The supplementary U.S. analysis uses Consumer Price Index (CPI) data and is evaluated over January 2022–September 2025. It has no equivalent separate validation stage, so its findings are exploratory.

The European and U.S. exercises use different inflation measures, news sources and predictor sets. Their error values should therefore be interpreted within each exercise rather than compared directly. Each corpus represents one English-language publication. The use of archived historical series and selected news fields also limits conclusions about operational real-time performance and broader news coverage.

Chapter 2 reviews the related literature. Chapter 3 describes the data and indicator construction, and Chapter 4 specifies the models and evaluation design. Chapter 5 presents and discusses the European results, supplementary U.S. findings, interpretation, limitations and dashboard workflow. Chapter 6 answers the research questions and identifies future research. Appendix A provides extended results and numerical diagnostics.

# Chapter 2 Literature Review

Regime-switching models and news-based indicators address different aspects of inflation prediction: the first allows dynamics to change, while the second broadens the information available to the model. The literature review brings these strands together, covering financial sentiment and empirical applications of text-based inflation prediction. It provides the basis for the EU-27 comparisons developed in this thesis.

## 2.1 Inflation Forecasting and Regime-Switching Models

Inflation forecasting becomes particularly difficult when economic conditions change abruptly. The post-pandemic surge illustrates this problem: examining Eurosystem and ECB staff projections for 2021 and early 2022, Chahad et al. (2022) [6] traced much of the underestimation to unexpected energy-price increases, supply bottlenecks and reopening effects. Energy-price conditioning assumptions were a major source of projection error.

These shifts provide a reason to consider models whose parameters can change across regimes. In earlier work on U.S. quarterly GDP-deflator inflation over 1960Q1–2003Q1, Binner et al. (2006) [7] compared linear AR, recurrent neural networks and a two-state Markov-switching AR model, reserving the final 100 observations for evaluation. The switching model allowed the intercept and innovation variance to vary across states while keeping the AR coefficients common. The nonlinear models performed similarly at one- and two-quarter horizons and generally outperformed linear AR; Markov switching performed best at three and four quarters. Whether this advantage extends to monthly EU-27 HICP, and how it compares with Random Forest and gradient boosting, motivates the first set of experiments in this thesis.

## 2.2 Financial Sentiment Analysis and Topic-Based Indicators

Araci (2019) [3] introduced FinBERT, a BERT-based model adapted to financial text through further pretraining and fine-tuning. The study evaluated financial sentiment classification on Financial PhraseBank and scoring on the Financial Question Answering (FiQA) dataset. Ten-fold cross-validation on the full Financial PhraseBank dataset yielded accuracy of 0.86 and an F1 score of 0.84. In this thesis, FinBERT provides sentiment scores for selected news texts; the inflation models then assess their predictive contribution.

Tilly and Livan (2021) [8] combined GDELT news topics and tone with market variables for eight countries. They reduced the news features using partial least squares and evaluated classifiers with walk-forward validation. Their target was the direction of five-day changes in ten-year breakeven inflation rates, predicted one day ahead. Adding news improved F1 scores in seven countries, with Mexico the exception; logistic regression and XGBoost were

strong performers. Although the target differs from realised consumer-price inflation, the study motivates examining topic and tone together.

The news indicators in this thesis cover broader economic developments, including energy, supply chains and geopolitical events. An article need not explicitly mention inflation to contribute to an indicator. Unlike the inflation-pressure classifiers reviewed below, the approach measures news tone and topic-specific negative intensity, then estimates their relationship with inflation empirically. Chapter 3 details their construction.

### 2.3 News-Based Inflation Forecasting and Nowcasting

The contribution of text to inflation prediction varies considerably across studies, with the source, target and forecasting horizon shaping the results. Eugster and Uhl (2024) [4] used roughly 730,000 news articles from 2003Q1 to 2021Q4 to predict U.S. inflation up to eight quarters ahead. Their sentiment-based predictions reduced pseudo-out-of-sample RMSE by about 30% relative to a naive random walk, depending on the horizon, and outperformed macroeconomic and Michigan survey benchmarks at shorter horizons. For the euro area, Boss et al. (2025) [9] reported substantial gains in a preprint using LLaMA-3-70B-Instruct to classify inflation pressure in Reddit submissions and comments. Over January 2018–December 2023, their best mixed-frequency specifications reduced RMSFE relative to an AR(1) benchmark by 27.7% for headline HICP and 32.1% for food inflation, with significance at the 5% and 1% levels, respectively.

The evidence is less favourable in other settings. Rambaccussing and Kwiatkowski (2020) [5] found that sentiment from U.K. newspapers helped predict output and unemployment but did not improve inflation nowcasts or short-term forecasts; performance also differed across publications and indicator constructions. In a more recent U.S. preprint, Allard et al. (2024) [10] fine-tuned a financial language model to classify news as indicating rising, falling or neutral inflation pressure. Adding the resulting monthly NEWS index to a Cleveland Fed-based nowcasting framework lowered RMSE from 0.0409 to 0.0407 over January 2020–December 2023. The Giacomini–White test returned $p = 0.19$, so this small reduction was not statistically significant at conventional levels.

These contrasting findings suggest that the usefulness of news depends on how the indicators are constructed and combined with other predictors. This thesis therefore tests general FinBERT sentiment alongside topic-specific negative-news indicators for EU-27 month-on-month HICP. Comparing inflation-only, macroeconomic and combined macroeconomic-news inputs across four model families separates the contribution of news from the choice of model. Validation and final-test results then show whether any gains persist across periods.

# Chapter 3 Data and Indicator Construction

The empirical analysis begins with two monthly datasets spanning January 2007–December 2025. The European application combines Eurostat series with Guardian news; the supplementary U.S. application pairs Federal Reserve Economic Data (FRED) series with New York Times articles. Each corpus is processed separately. The following sections define the inflation targets and macroeconomic predictors, explain how articles become monthly news indicators, and set out the calendar alignment used in evaluation. Transformations, lags and missing observations determine the sample available to each model.

### 3.1 European and U.S. Data Sources and Inflation Measures

The European dataset covers 228 calendar months. All macroeconomic series use Eurostat's EU27_2020 geography, which applies the fixed 27-country composition defined in 2020 throughout the historical period. The aggregate therefore differs from both the euro area and EU membership at each historical date. Eurostat's published aggregate series are used directly [1, 11].

The principal price series is the all-items Harmonised Index of Consumer Prices (HICP). The source series is not officially seasonally adjusted. The inflation target is its arithmetic month-on-month percentage change:

$$\pi_t = 100\left(\frac{P_t}{P_{t-1}} - 1\right)$$

Here, P□ is the HICP index for reference month t. Monthly inflation is measured in percent per month and is not annualised. Because January 2007 has no preceding price observation in the extracted sample, the dataset contains 227 monthly inflation values, from February 2007 to December 2025.

The U.S. target uses the seasonally adjusted all-items Consumer Price Index for All Urban Consumers (CPIAUCSL), produced by the Bureau of Labor Statistics and retrieved through FRED [12, 13]. Monthly inflation is calculated using the same arithmetic percentage-change formula as for HICP.

The U.S. dataset also covers 228 calendar months. October 2025 CPI is missing, preventing calculation of monthly inflation for October and November; December inflation can be calculated, but its first lag remains unavailable. There are consequently 225 monthly inflation observations in the full dataset. Although the analysis period extends to November 2025, the common evaluation sample ends in September and contains 45 months from January 2022.

### 3.2 Macroeconomic Predictors

The European predictors cover four economic channels: consumer energy prices, producer energy prices, labour-market conditions and industrial activity. Each series has monthly frequency and uses the same EU27_2020 geography as the target [11]. Index growth is measured in percent; changes in rates are measured in percentage points.

Consumer energy prices are measured by the energy component of HICP, and producer energy prices by the domestic energy producer price index (PPI). The PPI series covers the energy main industrial grouping and is neither seasonally nor calendar adjusted. Both enter the models as arithmetic monthly percentage changes, capturing price movements at different stages of the supply chain.

The seasonally adjusted unemployment rate uses Eurostat's total-sex and total-age categories and measures unemployment as a percentage of the labour force. Its first monthly difference enters the models in percentage points.

The industrial production volume index covers mining and quarrying, manufacturing, and electricity, gas, steam and air-conditioning supply. It is seasonally and calendar adjusted and enters as an arithmetic monthly percentage change.

Each transformed European macroeconomic predictor is evaluated at lags of one to six calendar months. A first lag pairs the observation for month $t - 1$ with the target for month t. Lags are calculated on the complete monthly calendar before incomplete observations are excluded, preserving their calendar meaning.

The U.S. predictors are the unemployment rate (UNRATE), effective federal funds rate (FEDFUNDS), industrial production index (INDPRO), and monthly West Texas Intermediate (WTI) spot oil price (MCOILWTICO), all retrieved through FRED [13]. Unemployment and the effective federal funds rate enter as first differences in percentage points. FEDFUNDS is the monthly average of daily effective rates, distinct from the Federal Open Market Committee's target range. Industrial production enters as an arithmetic monthly percentage change.

WTI is measured in U.S. dollars per barrel and enters as 100 times the first difference of its natural logarithm. Non-positive prices are treated as missing before taking logarithms. Predictor lags depend on the experiment. The U.S. variables represent economic channels related to those in Europe, but the two predictor sets are not identical.

### 3.3 News Processing and Monthly Sentiment Indicators

The European news corpus is collected through The Guardian Open Platform application programming interface (API) [14]. For each calendar month, the routine searches the Business and World sections together using the fixed query "(eurozone OR Europe) AND

(inflation OR oil OR energy OR unemployment OR war OR sanctions OR supply-chain OR shipping)" and retrieves all result pages. After duplicate article identifiers are removed, the resulting corpus contains 29,080 articles over 2007–2025. Publication dates determine the monthly assignment. The headline and short summary are cleaned by removing Hypertext Markup Language (HTML) markup, decoding HTML entities and normalising whitespace.

For Guardian articles, FinBERT receives the headline followed by the API's short-summary field, trailText. Full article bodies and keyword tags are excluded from this input; tags are retained only as metadata.

The Guardian query defines which articles enter the corpus. Its geographical terms can admit coverage outside the EU-27 and miss relevant country-specific articles that lack those terms. The resulting sample represents the selected English-language coverage of one publication, a limitation considered when interpreting the news indicators.

The supplementary corpus contains 18,063 selected New York Times articles from 2007–2025, retrieved through the Archive API [15]. An article is retained when its combined text and editorial metadata contain at least one predefined macroeconomic term and one predefined U.S.-related term. The searchable fields are the headline, abstract, lead paragraph, snippet, keyword values, news desk and section name. Matching uses case-insensitive substrings. Records with invalid publication dates or empty headlines are excluded, and duplicate identifiers and date–headline pairs are removed.

FinBERT receives the available headline, abstract, lead paragraph and snippet, with identical repeated fragments removed. Full article bodies are excluded. Differences in selection rules and input fields limit direct comparisons of sentiment levels between the New York Times and Guardian corpora.

**FinBERT scoring**

Bidirectional Encoder Representations from Transformers (BERT) represents words in context by attending to surrounding tokens [16]. For sequence classification, a classifier uses the representation associated with a special classification token to assign a label to the input text. This contextual approach can distinguish expressions that an isolated keyword count would treat alike.

Sentiment is scored with the existing ProsusAI/finbert checkpoint associated with Araci's adaptation of BERT to financial text [3, 17]. The model documentation describes additional language-model training on a subset of Reuters TRC2 and supervised sentiment training on Financial PhraseBank [17]. This financial focus provides the rationale for using the checkpoint on the selected news articles.

The checkpoint has 12 encoder layers, 12 attention heads per layer and a hidden dimension of 768 [17]. Its classification layer produces scores for positive, negative and neutral financial sentiment, which softmax converts to three class probabilities.

FinBERT is applied in inference mode, without further fine-tuning. Each article's combined text forms one sequence, padded as needed and truncated to a maximum of 512 tokens, including special tokens. Classification therefore depends on the text within that limit. The resulting measures of financial tone enter the inflation models as additional predictors.

**Monthly sentiment indicators**

For article i, let $p_i^+$ and $p_i^-$ denote the model's positive and negative class probabilities. The signed sentiment score is:

$$s_i = p_i(\text{positive}) - p_i(\text{negative})$$

The score ranges from −1 to 1. The principal monthly indicator is the unweighted mean of the N□ selected article scores: S□ = (1/N□)$\Sigma_i s_i$. Each article has equal weight within its month.

The dataset also retains the median score, sample standard deviation, distinct article count, and shares of positive, neutral and negative labels. The predicted label is the class with the highest probability. Thus, the negative-label share is the proportion of articles classified as negative, rather than the average negative probability. The standard deviation measures differences in signed tone across articles, rather than uncertainty about each classification.

The topic indicators combine dictionary matches with FinBERT negative-class probabilities. The three dictionaries use the following exact search terms:

**Geopolitical:** war; armed conflict; invasion; military; missile; airstrike; attack; sanction; embargo; blockade; geopolitical; terrorism.

**Energy:** oil; crude; petroleum; gasoline; diesel; natural gas; fuel; refinery; pipeline; opec; energy supply; energy prices; electricity.

**Supply chain:** supply chain; shipping; freight; port; logistics; container; cargo; trade route; semiconductor; chip shortage; suez; red sea; bottleneck.

An article matches a topic when its sentiment-input text contains any dictionary term, using case-insensitive substring matching. Articles can match several topics. It also examines the full stored input text, whereas FinBERT is limited to 512 tokens.

For topic k and reference month t, the negative-intensity indicator is defined as:

$$R_{k,t} = \frac{1}{N_t} \sum_{i=1}^{N_t} I_{i,k}\, p_i(\text{negative})$$

N□ counts all selected articles in month t; $p_i^-$ is article i's negative class probability; and $I_i$□ is one when the stored text matches topic k and zero otherwise. The denominator is the whole monthly corpus, not only the matching articles.

The indicator ranges from 0 to 1 and combines topic prevalence with negative financial tone. When a topic has at least one match, its value equals the share of matching articles multiplied by their mean negative probability. A zero is a valid observed value: matching articles contribute no negative-probability weight. Overlapping dictionaries mean that the three indicators cannot be interpreted as mutually exclusive components of total risk.

Negative financial tone can accompany weak demand or disrupted energy supply, with different implications for inflation. The topic measures are therefore interpreted as negative-news intensity within each topic. Their predictive contribution is assessed empirically in Chapter 4.

The predictive comparisons in Chapter 4 examine whether these article-level classifications and topic matches provide useful information for monthly inflation estimation.

### 3.4 Data Alignment, Information Availability and Descriptive Statistics

Macroeconomic observations and news indicators are merged by reference month on a regular monthly calendar. Predictor timing depends on the experiment. The initial European forecasting comparisons use prior-month news. The European sentiment-nowcast and final combined comparisons instead use news from target month t together with lagged inflation and macroeconomic predictors. Using target-month news defines a retrospective month-end nowcast, with a later information set than a forecast using only prior-month news.

The supplementary U.S. analysis has the same main objective: to use news published during a month to estimate inflation for that same month. The generic sentiment extensions therefore use same-month news indicators. The project also tests lagged news as alternative inputs: previous-month sentiment in the AR-HMM benchmark, sentiment lags of one to six months in LASSO, and previous-month topic indicators in the subject-specific comparisons. These additional tests assess whether earlier news also contains useful predictive information.

Missing macroeconomic observations are retained as gaps. Monthly changes and lags are computed on the complete calendar grid before incomplete rows are dropped, ensuring that a one-month lag always refers to the preceding calendar month, including when gaps occur. An observation is excluded from an experiment only after the transformations have been calculated and a required target or predictor is found to be unavailable.

Coverage checks require all five European source series and monthly Guardian sentiment to span the full 228-month calendar. U.S. news indicators also cover all 228 months, although gaps in macroeconomic data reduce the usable sample. Transformations and predictor requirements further determine eligibility. Direct model comparisons use common months.

European seasonal specifications centre inflation using calendar-month means estimated from the relevant training period. Each January observation, for example, has the training sample's January mean subtracted. This removes the estimated monthly mean, including its average inflation component. The procedure is a model-specific calendar adjustment, distinct from Eurostat's official seasonal adjustment.

The same training-derived means are applied to the subsequent evaluation block and added back to predictions before errors are calculated on the original inflation scale. Where predictors are standardised, their means and sample standard deviations are also estimated from training data. These quantities are recomputed for each validation fold and remain fixed within its evaluation block.

European structure selection uses the 2017–2021 validation period. Later comparisons use three validation blocks—2017–2018, 2019–2020 and 2021—with expanding training periods. The final European test contains 48 months, from January 2022 to December 2025. Chapter 4 sets out the model estimation and selection procedures.

Figure 3.1 plots the observed monthly inflation target before model-specific calendar-month centring and shows the pronounced increase during 2022.

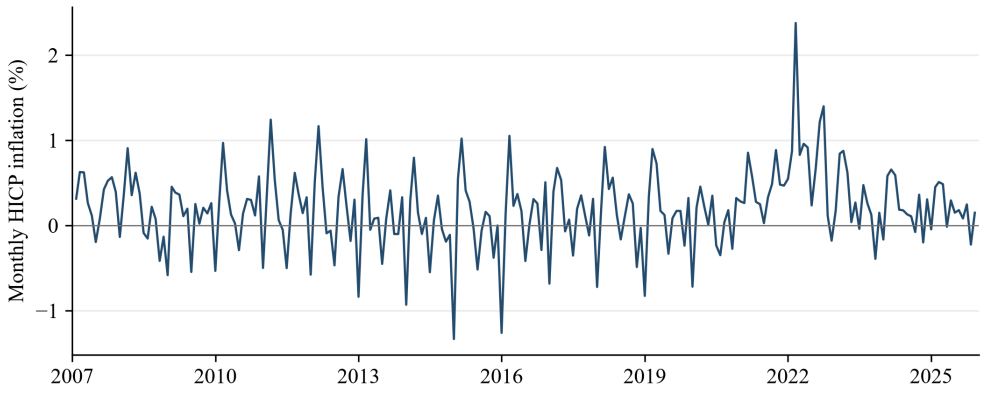



Figure 3.1. EU-27 month-on-month HICP inflation, February 2007–December 2025

Across 228 months, Guardian coverage averages 127.5 articles per month and ranges from 44 to 320. The average monthly mean sentiment is −0.284, and the average monthly negative-label share is 52.4%. The corresponding monthly averages of geopolitical, energy and supply-chain negative intensity are 0.133, 0.037 and 0.066. Each month has equal weight in these summaries; they are not pooled article-level averages.

Differences between the topic indicators reflect dictionary coverage, corpus composition and financial tone.

Figure 3.2 shows how the number of selected Guardian articles varies over time.

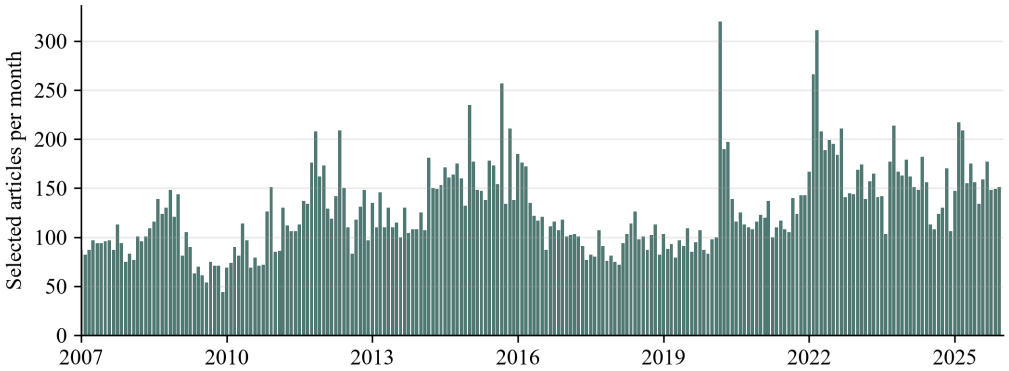



Figure 3.2. Monthly number of selected Guardian articles, January 2007–December 2025

Figure 3.3 shows the monthly mean sentiment and topic-specific negative-intensity indicators, with a separate vertical scale for each panel. The series describes variation within the selected corpus.

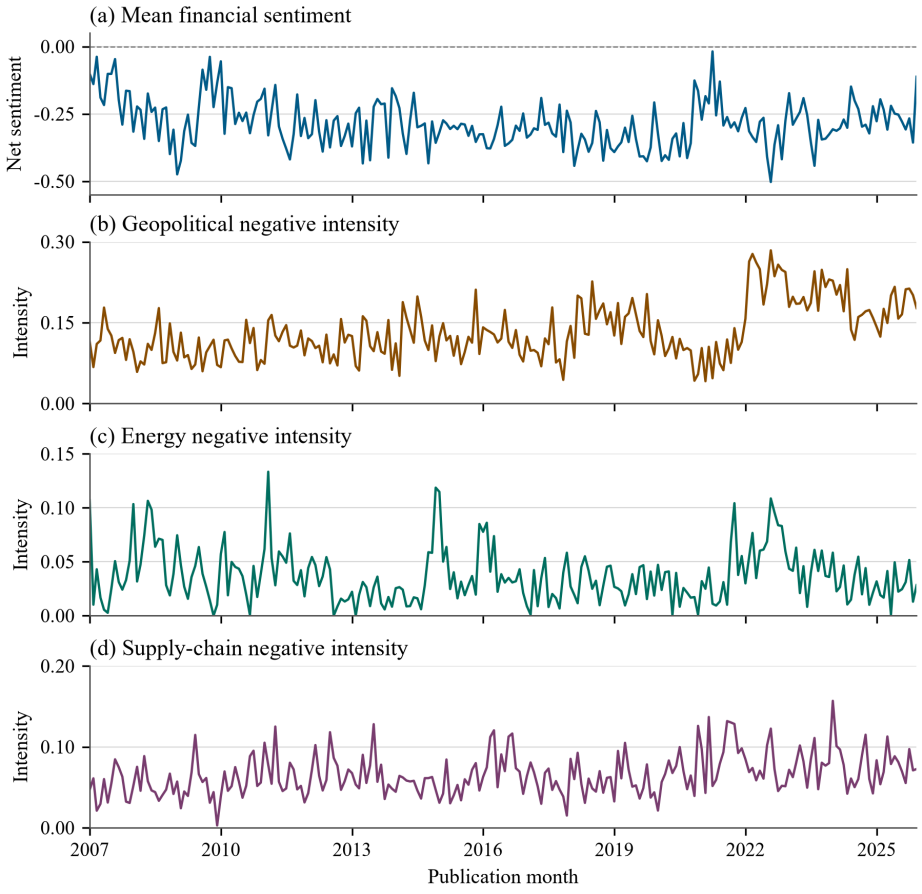



Figure 3.3. Monthly Guardian sentiment and topic-risk indicators, January 2007–December 2025

# Chapter 4 Models and Evaluation Design

The model comparisons separate the contribution of regime switching from that of additional information. European validation first selects an AR-HMM structure. Subsequent experiments add macroeconomic and news predictors, with linear AR, Random Forest and gradient boosting as benchmarks. The supplementary U.S. analysis follows the same sequence of predictor comparisons.

## 4.1 Benchmarks: AR, Random Forest and Gradient Boosting

The autoregressive model, AR(p), estimates inflation using its previous p monthly observations. In the equation below, $y_t$ is the inflation target and $y_{t-j}$ is inflation j months earlier. The initial model-selection experiment uses observed monthly inflation; subsequent European comparisons use the calendar-month-centred target defined in Chapter 3.

$$y_t = c + \sum_{j=1}^{p} \phi_j y_{t-j} + \varepsilon_t$$

The intercept c represents the baseline level, $\varphi_j$ measures the contribution of each inflation lag, and $\varepsilon_t$ is the unexplained error. Ordinary least squares estimates the coefficients. AR(1) uses only the previous month. The initial comparison also includes a simple benchmark that repeats the previous month's inflation.

Random Forest and gradient boosting estimate inflation directly from the supplied predictors. They are fitted separately from AR-HMM and do not use its hidden states or state probabilities. The four model families receive the same observed inputs in the matched European comparison. The tree ensembles are implemented in Scikit-learn [18].

Random Forest averages predictions from multiple regression trees, each fitted to a resampled training dataset [18]. Gradient boosting builds trees sequentially, with each new tree correcting errors left by the preceding trees [18]. These models provide nonlinear alternatives to linear AR.

Three manually specified configurations are compared within each tree-model family. The candidates provide a limited sensitivity analysis of model flexibility and regularisation. Their numerical settings are treated as alternatives for validation, rather than as established optimal values. The configuration with the lowest validation RMSE is retained.

**Random Forest RF1**: n_estimators = 300; max_depth = 4; min_samples_leaf = 2; max_features = 0.7.

**Random Forest RF2**: n_estimators = 300; max_depth = None (no explicit depth limit); min_samples_leaf = 4; max_features = 0.7.

**Random Forest RF3**: n_estimators = 300; max_depth = 6; min_samples_leaf = 6; max_features = 1.0.

**Gradient Boosting GB1**: n_estimators = 100; learning_rate = 0.03; max_depth = 2; min_samples_leaf = 4; subsample = 0.8.

**Gradient Boosting GB2**: n_estimators = 100; learning_rate = 0.05; max_depth = 2; min_samples_leaf = 6; subsample = 0.8.

**Gradient Boosting GB3**: n_estimators = 150; learning_rate = 0.03; max_depth = 1; min_samples_leaf = 4; subsample = 1.0.

The Random Forest candidates hold the ensemble size at 300 trees while varying depth, minimum leaf size and the fraction of features considered at each split. They contrast depth-limited trees with an unrestricted-depth alternative that still requires multiple observations per leaf [18]. The boosting candidates use shallow trees and small learning rates, varying the number of boosting stages, leaf size and observation subsampling; GB3 uses depth-1 trees (stumps), while GB1 and GB2 allow depth-2 trees [18].

All six configurations use random_state = 42. Within each model family and predictor group, the candidate with the lowest pooled validation RMSE over the 2017–2021 blocks in Section 4.4 is retained, refitted through December 2021 and evaluated over 2022–2025. The reported ranking is conditional on this small candidate set.

### 4.2 AR-HMM and Alternative Switching Structures

AR-HMM allows the relationship between past and current inflation to change between unobserved states. A Markov transition model describes the probability of remaining in a state or moving to another state. The switching models are estimated with Statsmodels [19]. For the selected two-state model with one inflation lag, the inflation equation is:

$$y_t = c_{S_t} + \varphi_{S_t} y_{t-1} + \varepsilon_t, \qquad S_t \in \{1, 2\}$$

$S_t$ is the state in month t, either 1 or 2. Each state has its own intercept c and lag coefficient φ: state 1 uses $c_1$ and $\varphi_1$, while state 2 uses $c_2$ and $\varphi_2$. The error $\varepsilon_t$ has a common variance in the European model [19].

The initial European experiment compares one, two and three inflation lags, with two or three states for AR-HMM. Candidates are fitted through December 2016 and evaluated over 2017–2021 using observed monthly inflation. Two further variants restrict either the autoregressive coefficient or the mean relationship to be common across states. Table 4.1 reports this comparison.

Table 4.1. European model-structure validation, 2017–2021. Fitting ends in December 2016. RMSE and MAE are in percentage points of monthly inflation.

| Specification | RMSE | MAE |
|---|---|---|
| AR(1) | 0.3895 | 0.2987 |
| AR-HMM(1), 2 states | 0.3775 | 0.2900 |
| AR-HMM(1), 3 states | 0.3844 | 0.2937 |
| AR(2) | 0.3997 | 0.3030 |
| AR-HMM(2), 2 states | 0.3898 | 0.3054 |
| AR-HMM(2), 3 states | 0.3813 | 0.2855 |
| AR(3) | 0.3983 | 0.3029 |
| AR-HMM(3), 2 states | 0.4271 | 0.3433 |
| AR-HMM(3), 3 states | 0.3963 | 0.3034 |
| AR-HMM(1), 2 states, common AR | 0.3862 | 0.2950 |
| MS-Variance AR(1), 2 states | 0.3932 | 0.2916 |

The two-state AR-HMM(1) has the lowest validation RMSE among the switching candidates, 0.3775, and is retained for the main European experiments. Selection is based on RMSE; the three-state AR-HMM(2) has a slightly lower MAE. Figure 4.1 shows the state-and-lag comparison.

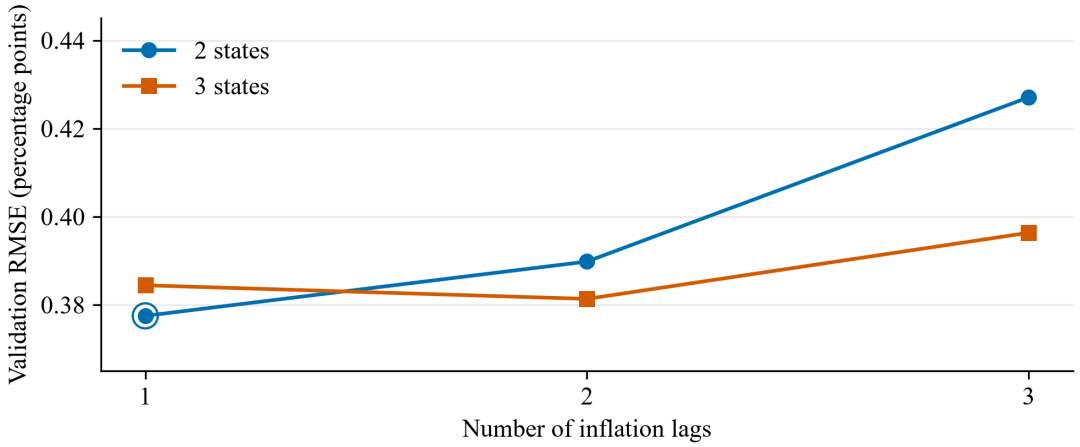



Figure 4.1. Validation RMSE by number of inflation lags and latent states, 2017–2021. The circled point identifies the selected two-state AR-HMM(1).

For each month, the fitted AR-HMM produces an estimate from each state and averages them using predicted state probabilities. These probabilities are calculated before observing that month's inflation. Once inflation is observed, the probabilities are updated for the next month; the fitted coefficients remain fixed within the evaluation period.

Figure 4.2 shows this process for the European model fitted through 2021. The upper panel compares observed inflation with model estimates and marks the most probable state. The lower panel shows both predicted state probabilities. State labels describe the fitted inflation relationships.

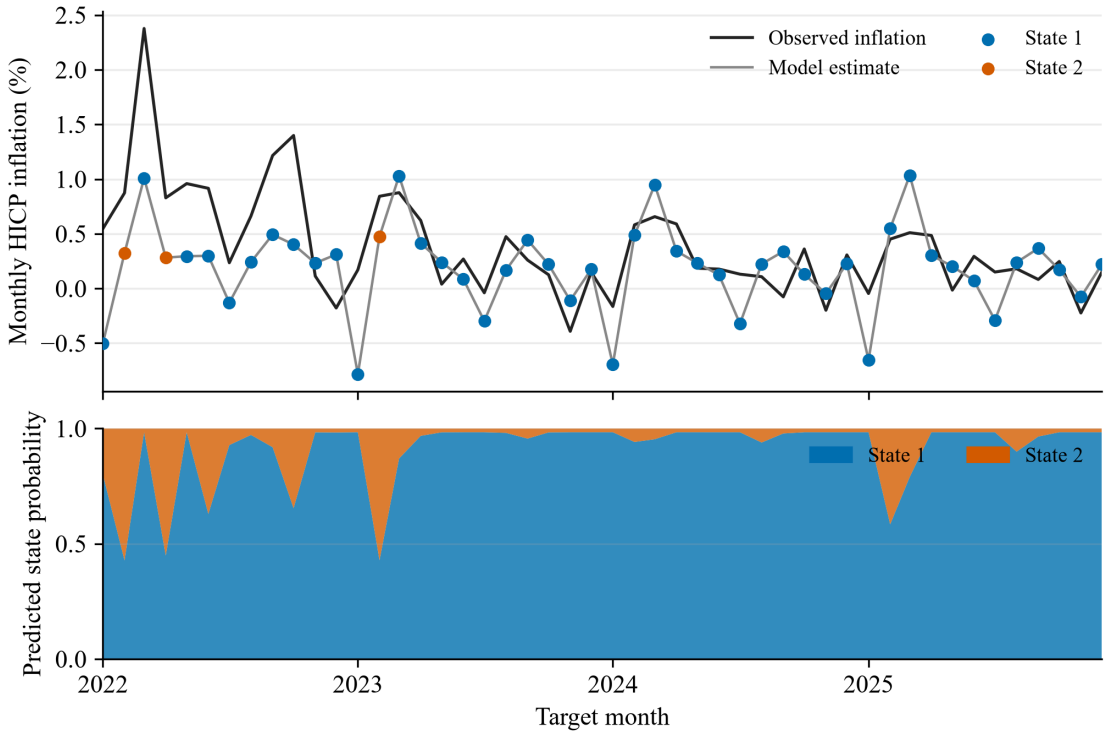



Figure 4.2. European two-state AR-HMM(1), 2022–2025. Top: monthly inflation, model estimates and predicted states. Bottom: state probabilities formed before observing each month's inflation. Calendar effects are restored to the estimates.

### 4.3 Predictor Groups and Experimental Comparisons

Additional predictors enter the inflation equation through $x_t$. Their coefficients can differ between AR-HMM states; the linear model uses common coefficients:

$$y_t = c_{S_t} + \sum_{j=1}^{p} \phi_{j,S_t} y_{t-j} + \boldsymbol{\beta}'_{S_t} \boldsymbol{x}_t + \varepsilon_t$$

The vector $x_t$ contains the inputs for target month t. Macroeconomic variables are lagged, while the principal news specifications use articles from the target month. They therefore estimate that month's inflation at month-end.

Before the later European comparisons, calendar-month means are subtracted from inflation using training data only. These effects are added back before errors are calculated. Predictor scaling also uses training data only. Chapter 3 describes these transformations.

The macroeconomic experiments examine unemployment changes, industrial-production growth, energy HICP inflation and energy PPI growth. After an initial lag-1 comparison, lags 1–6 are assessed separately for each indicator using validation RMSE. Table 4.2 gives the selected lags; inflation itself remains at lag 1.

Table 4.2. Selected European macroeconomic predictor lags. Each indicator is assessed at lags 1–6 using combined validation RMSE with the calendar-month-centred two-state AR-HMM(1).

| Indicator | Candidate lags | Selected lag (months) |
|---|---|---|
| Unemployment | 1–6 | 3 |
| Industrial Production | 1–6 | 1 |
| Energy HICP | 1–6 | 1 |
| Energy PPI | 1–6 | 1 |

The news experiments compare mean and median sentiment, negative share, dispersion and the three topic-risk indicators. A focused energy comparison tests whether current-month energy-risk news adds information beyond energy PPI growth at lag 1: it compares inflation only, macro only, news only and their combination.

The matched four-family comparison uses the three predictor groups in Table 4.3. It starts with inflation lag 1, adds the four macroeconomic inputs in Table 4.2, then adds current-month mean sentiment and geopolitical, energy and supply-chain indicators. Each group is supplied to every model family.

Table 4.3. Common predictor groups in the European four-family comparison. All groups use the calendar-month-centred inflation target and its first lag.

| Group | Inputs supplied for target month t | Model families |
|---|---|---|
| Inflation only | Centred inflation, lag 1 | Linear AR; AR-HMM; Random Forest; gradient boosting |
| Inflation + macro | Inflation lag 1; unemployment change lag 3; industrial-production growth, energy HICP inflation and energy PPI growth, each lag 1 | Same four families |
| Inflation + macro + news | All preceding inputs; target-month mean sentiment and geopolitical, energy and supply-chain indicators | Same four families |

Comparing families within a group assesses model performance with the same inputs. Comparing predictor groups within a family measures the contribution of macroeconomic information and then news. The macroeconomic lags selected in the AR-HMM experiment are shared across all four families. This keeps the information sets comparable, but the lags were selected using AR-HMM rather than tuned separately for the tree models. Consequently, the comparison evaluates performance under shared predictors and the limited hyperparameter candidates in Section 4.1.

### 4.4 European Validation and U.S. Evaluation Design

The European evaluation separates model selection from final assessment. Training uses earlier observations to estimate coefficients or fit trees. Validation measures performance on later observations and guides choices such as the number of states, lag order, predictor set or tree settings. The final test then assesses the chosen specification on 2022–2025, which is not used to make those choices.

**Stage 1: select the AR-HMM structure.** Each state-and-lag candidate is fitted using observations through December 2016 and evaluated over the full 2017–2021 validation period. This is one fitting exercise per candidate, followed by one five-year validation block. The selected structure has two states and one inflation lag.

**Stage 2: compare predictors and tree-model settings.** Later comparisons divide the same 2017–2021 validation period into three successive blocks. Each candidate is fitted three times: through 2016 for evaluation in 2017–2018, through 2018 for evaluation in 2019–2020, and through 2020 for evaluation in 2021. The training window expands because observations that were previously used for validation have become historical data by the next

fitting date. These three blocks belong to the later comparisons; they are not extra steps within the initial structure-selection exercise. Table 4.4 summarises both stages and the final refit.

Table 4.4. European evaluation schedule. Stage 1 selects the AR-HMM structure; Stage 2 compares later candidate specifications using three expanding-training validation blocks. Final assessment follows a refit through 2021.

| Stage / step | Training data | Validation data | Test data |
|---|---|---|---|
| 1. Structure selection | 2007–2016 | 2017–2021 | — |
| 2. Validation block 1 | 2007–2016 | 2017–2018 | — |
| 2. Validation block 2 | 2007–2018 | 2019–2020 | — |
| 2. Validation block 3 | 2007–2020 | 2021 | — |
| 3. Final refit and test | 2007–2021 | — | 2022–2025 |

**Stage 3: refit and test the selected specification.** Once validation has determined the settings, the model is fitted again using all eligible observations through December 2021. The settings remain fixed, but the coefficients or trees are re-estimated using the larger training sample. This final fit is evaluated over January 2022–December 2025.

The reuse of 2017–2021 is deliberate: those years first help select the settings, then contribute to estimating the final model. They are not an independent test sample. The separate 2022–2025 period provides the final assessment. Table 4.4 therefore shows a dash in the final row's validation column: model selection has already been completed. The training ranges indicate calendar coverage; lags and missing observations determine which months are eligible for fitting.

Within each validation or test block, coefficients or trees remain fixed while the monthly inputs advance. AR-HMM state probabilities are updated after each observed inflation outcome, so the state information changes even though the fitted coefficients do not. Training-derived calendar-month means and predictor-scaling quantities are also fixed within a block and re-estimated when the training window expands.

Figure 4.3 illustrates how the European sample is partitioned across the three stages: the initial structure-selection split (top row), the three expanding-window validation folds (middle rows), and the final refit through 2021 followed by testing over 2022–2025 (bottom row).

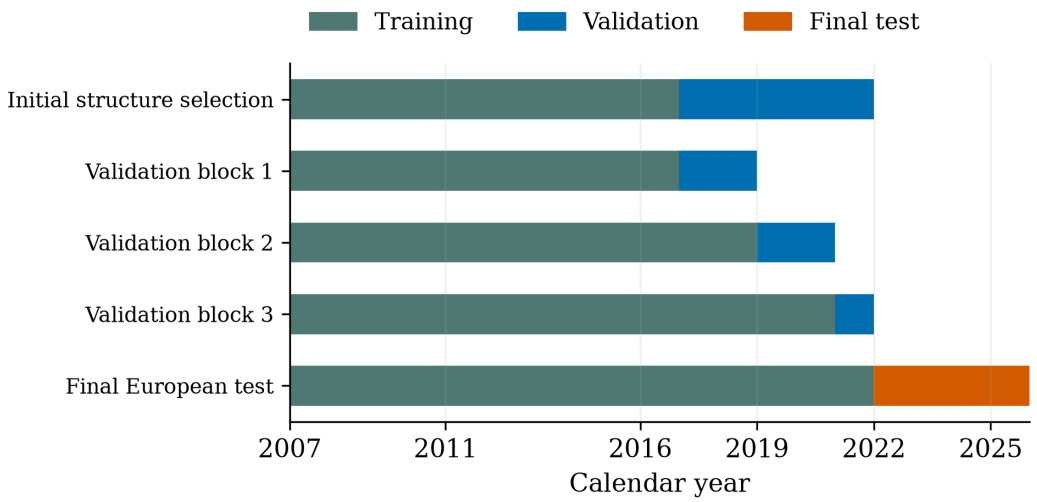



Figure 4.3. How the European sample is reused as time advances.

For each candidate in Stage 2, the three validation blocks produce 24, 24 and 12 monthly errors, respectively. These 60 errors are pooled to calculate a single validation RMSE. Candidate selection uses this pooled score, rather than a simple average of the three block RMSE values, so each scored month receives equal weight. The eligible configuration with the lowest pooled RMSE is retained for each model family and predictor group.

The U.S. comparison uses a common fitting sample from July 2008 to December 2021 and 45 test months from January 2022 to September 2025. It follows the inflation-only, macroeconomic and news comparison sequence. Because these specifications were compared on the test period without a separate validation stage, the U.S. findings are exploratory.

The U.S. analysis compares linear AR with two-state switching models, examining inflation lags 1–6 and the addition of macroeconomic and news inputs. Generic-news specifications include target-month sentiment, while the separate topic-risk comparison uses lagged indicators.

### 4.5 Evaluation Metrics and Fair Comparison

Errors are calculated on the original inflation scale. For each scored month, $e_t$ is observed inflation minus the model estimate; T is the number of common scored months. The accuracy measures are:

$$e_t = y_t - \hat{y}_t, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T} e_t^2}$$

$$\mathrm{MAE} = \frac{1}{T}\sum_{t=1}^{T} |e_t|, \qquad \mathrm{ME} = \frac{1}{T}\sum_{t=1}^{T} e_t$$

RMSE is the main selection measure and gives greater weight to large errors. MAE reports the average absolute error. Lower values indicate better accuracy; both are measured in percentage points. ME measures average bias: positive values indicate underestimation and negative values indicate overestimation.

Relative RMSE gain is $100 \times (\text{RMSE}_{\text{reference}} - \text{RMSE}_{\text{model}}) / \text{RMSE}_{\text{reference}}$. A positive gain means lower error than the stated reference model. Comparisons use the same target, input timing and scored months.

Predictive accuracy is assessed using exploratory, two-sided Diebold-Mariano tests [20, 21] on paired out-of-sample forecast errors. The loss function is the squared forecast error, $L_t = (y_t - \hat{y}_t)^2$, and the null hypothesis is equal expected loss between competing forecasts. The tests use Newey-West standard errors with three lags, the Harvey-Leybourne-Newbold small-sample correction (h = 1), and Holm adjustment across the two AR-HMM comparisons. Statistical significance is assessed at the 5% level.

# Chapter 5 Results and Discussion

The European results are organised around a practical question: what improves inflation estimates beyond the information in past inflation? The analysis moves from baseline models and calendar-month centring to macroeconomic inputs, news indicators and the matched comparison of model families. Validation covers 2017–2021 and final testing covers 2022–2025. Errors are reported in percentage points.

## 5.1 From Model Development to the Final Test

The main achievement is the full-information AR-HMM's final-test RMSE of 0.3959, the lowest among the 12 matched European specifications. Adding news reduces its error by 11.2% relative to macroeconomic inputs alone. Sections 5.2–5.5 establish the earlier experiments; Section 5.6 presents this result, followed by the supplementary U.S. analysis, interpretation and limitations.

## 5.2 European Baseline Performance

In the initial uncentred experiment, final-test RMSE falls from 0.5060 for the previous-month benchmark to 0.4984 for linear AR(1), 0.4829 for variance-only switching and 0.4658 for the mean-switching AR-HMM(1). Regime switching improves accuracy within this initial comparison.

The matched calendar-centred benchmark is closer: inflation-only linear AR and two-state AR-HMM both reach RMSE of 0.4331, while Random Forest and gradient boosting give 0.4466 and 0.4523 (Table 5.3). Figure 5.1 compares their estimated paths.

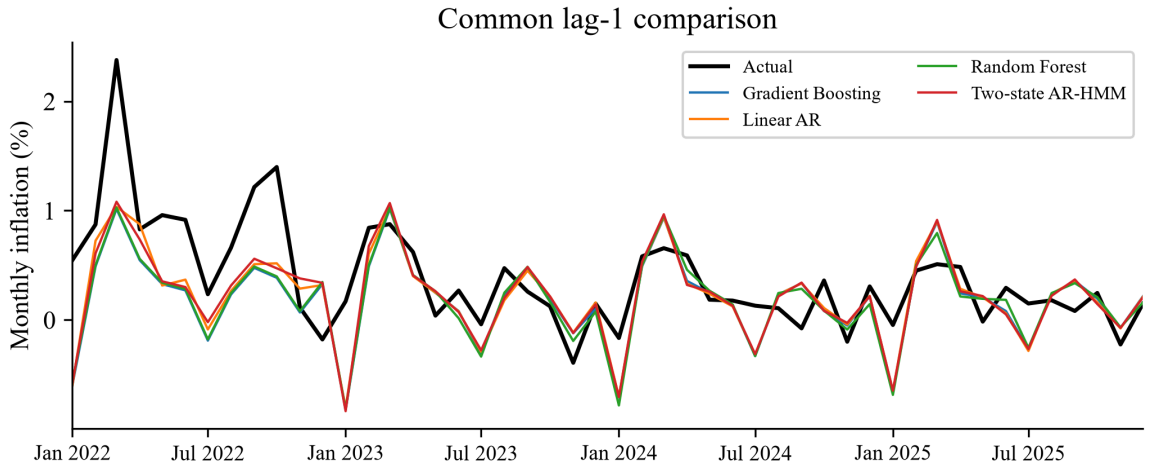



Figure 5.1. EU-27 observed monthly inflation and four model families using the same lag-1 information set, 2022–2025.

All four models underestimate the largest early-2022 increases. This shared difficulty motivates the subsequent tests of calendar treatment and additional information.

### 5.3 Calendar Treatment and Regime Interpretation

Chapter 4 selects two states and one inflation lag by validation RMSE: 0.3775, compared with 0.3813 for the nearest three-state, two-lag candidate. This two-state AR-HMM(1) is retained for the European predictor experiments.

In the separate seasonal stage, calendar-month centring lowers AR-HMM final-test RMSE from 0.4658 to 0.4553; adding lagged subject risks lowers it to 0.4396. Centred linear AR(1) gives 0.4331. Figure 5.2 illustrates the combined change in calendar treatment and predictors.

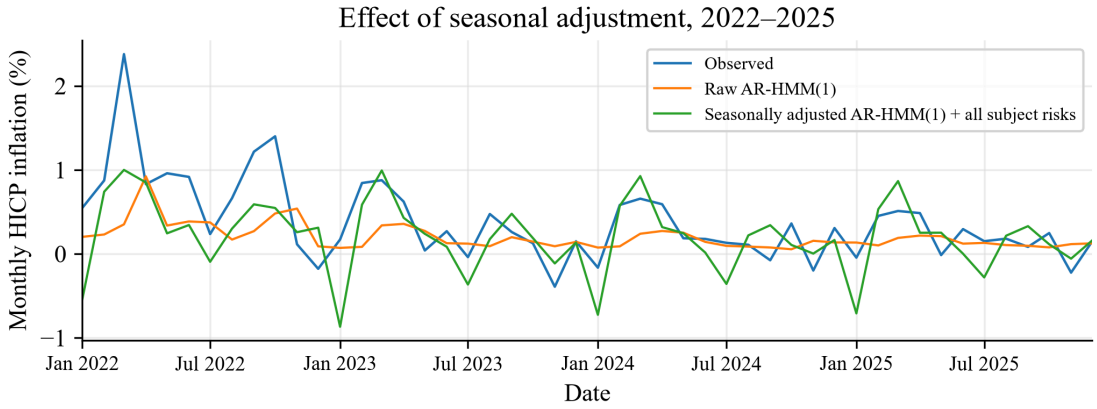



Figure 5.2. Uncentred AR-HMM and the seasonal-stage model with lagged subject risks, 2022–2025.

The European states describe different conditional inflation relationships with a common innovation variance. They should therefore be interpreted as fitted mean states, rather than low- and high-volatility regimes.

### 5.4 Macroeconomic Predictors

Table 5.1 summarises the macroeconomic additions to the calendar-centred AR-HMM(1). The fixed-lag experiment uses one-month lags; the last two rows report the separate validation-selected-lag experiment. Energy block combines energy HICP and PPI; slow macro combines unemployment and industrial production. Selected-lag order is unemployment, industry, energy HICP and energy PPI.

Table 5.1. European macroeconomic inputs: validation and final-test RMSE, in percentage points. Validation: 2017–2021; test: 2022–2025.

| Added inputs | Lag (months) | Validation RMSE | Test RMSE |
|---|---|---|---|
| Baseline | — | 0.2416 | 0.4331 |
| unemployment | 1 | 0.2414 | 0.4304 |
| industrial production | 1 | 0.2434 | 0.4342 |
| energy HICP | 1 | 0.2393 | 0.4674 |

| | | | |
|---|---|---|---|
| energy PPI | 1 | 0.2267 | 0.4594 |
| energy block | 1 | 0.2247 | 0.4538 |
| slow macro | 1 | 0.2506 | 0.4356 |
| all macro | 1 | 0.2335 | 0.4509 |
| Unemployment (selected lag) | 3 | 0.2365 | 0.4279 |
| All macro (selected lags) | 3, 1, 1, 1 | 0.2444 | 0.4647 |

Energy inputs give the strongest validation gains, while unemployment gives the lowest final-test error among the listed additions. Selecting unemployment lag 3 improves RMSE to 0.4279. Figure 5.3 compares the baseline and energy-block estimates during 2022, when both models miss the largest monthly increases.

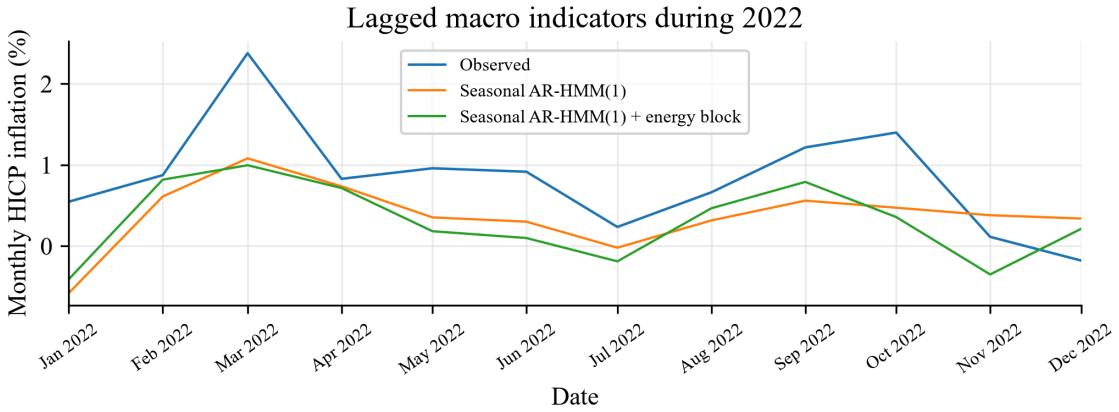



Figure 5.3. European observed inflation and AR-HMM estimates with and without the lagged energy block during 2022

### 5.5 News Sentiment and Subject-Specific Risk

Table 5.2 compares target-month Guardian indicators added to the inflation-lag baseline. These are month-end estimates using news published during the month being estimated.

Table 5.2. European news additions: validation and final-test RMSE, in percentage points. Calendar-centred AR-HMM(1); validation: 2017–2021; test: 2022–2025.

| Added news indicator | Validation RMSE | Test RMSE |
|---|---|---|
| Baseline | 0.2408 | 0.4331 |
| mean sentiment | 0.2405 | 0.4374 |
| median sentiment | 0.2434 | 0.4347 |
| negative share | 0.2462 | 0.4335 |
| sentiment dispersion | 0.2417 | 0.4383 |
| geopolitical risk | 0.2485 | 0.4477 |

| | | |
|---|---|---|
| energy risk | 0.2395 | 0.4337 |
| supply-chain risk | 0.2407 | 0.4263 |
| all subject risks | 0.2557 | 0.4156 |

Generic sentiment produces little change in final-test RMSE. The joint topic-risk model has the lowest test error in this experiment, at 0.4156, approximately 4.0% below baseline; energy risk has the lowest validation error.

The focused energy-PPI and energy-risk experiment gives test RMSE of 0.4331 for baseline, 0.4594 for macro only, 0.4337 for news only and 0.4743 for both (Figure 5.4; Table A.4). This narrower combination does not improve accuracy.

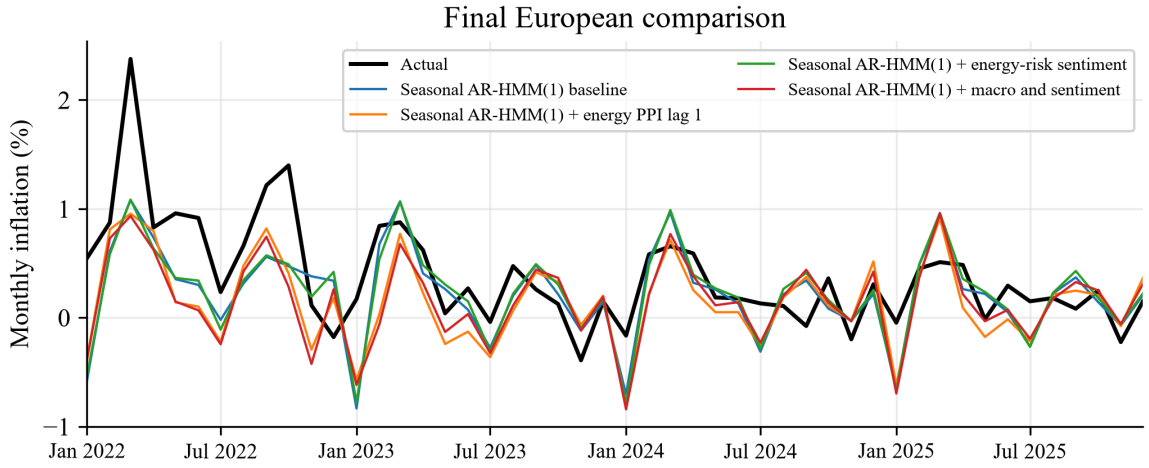



Figure 5.4. Focused European baseline, energy-PPI, energy-risk and combined estimates, 2022–2025.

### 5.6 Matched Comparison and the Main Final-Test Result

Table 5.3 compares four independent model families using the same inflation-only, macroeconomic, and macroeconomic-plus-news information sets. Within each group, the predictors, calendar treatment and scored months are shared. Figure 5.5 summarises the validation and final-test rankings.

Table 5.3. Matched European RMSE by model family and information set, in percentage points. Validation: 2017–2021 (60 months); final test: 2022–2025 (48 months), also divided into 2022 and 2023–2025.

| Inputs / model | Validation | 2022–2025 | 2022 | 2023–2025 |
|---|---|---|---|---|
| Inflation / Linear AR | 0.2414 | 0.4331 | 0.6872 | 0.3044 |
| Inflation / AR-HMM | 0.2416 | 0.4331 | 0.6820 | 0.3083 |
| Inflation / Random Forest | 0.2515 | 0.4466 | 0.7193 | 0.3057 |
| Inflation / Gradient Boosting | 0.2441 | 0.4523 | 0.7302 | 0.3084 |

| | | | | |
|---|---|---|---|---|
| + macro / Linear AR | 0.2357 | 0.4585 | 0.7051 | 0.3385 |
| + macro / AR-HMM | 0.2742 | 0.4460 | 0.6686 | 0.3409 |
| + macro / Random Forest | 0.2533 | 0.4622 | 0.7323 | 0.3257 |
| + macro / Gradient Boosting | 0.2470 | 0.4637 | 0.7385 | 0.3238 |
| + macro + news / Linear AR | 0.2356 | 0.4556 | 0.7299 | 0.3149 |
| + macro + news / AR-HMM | 0.2432 | 0.3959 | 0.5960 | 0.3009 |
| + macro + news / Random Forest | 0.2476 | 0.4615 | 0.7594 | 0.3029 |
| + macro + news / Gradient Boosting | 0.2446 | 0.4665 | 0.7831 | 0.2928 |

Adding news to the AR-HMM with inflation and macroeconomic predictors reduced RMSE by 11.25% over 2022-2025. An exploratory two-sided Diebold-Mariano test [20, 21] found this improvement statistically significant at the 5% level (Holm-adjusted p = 0.018).

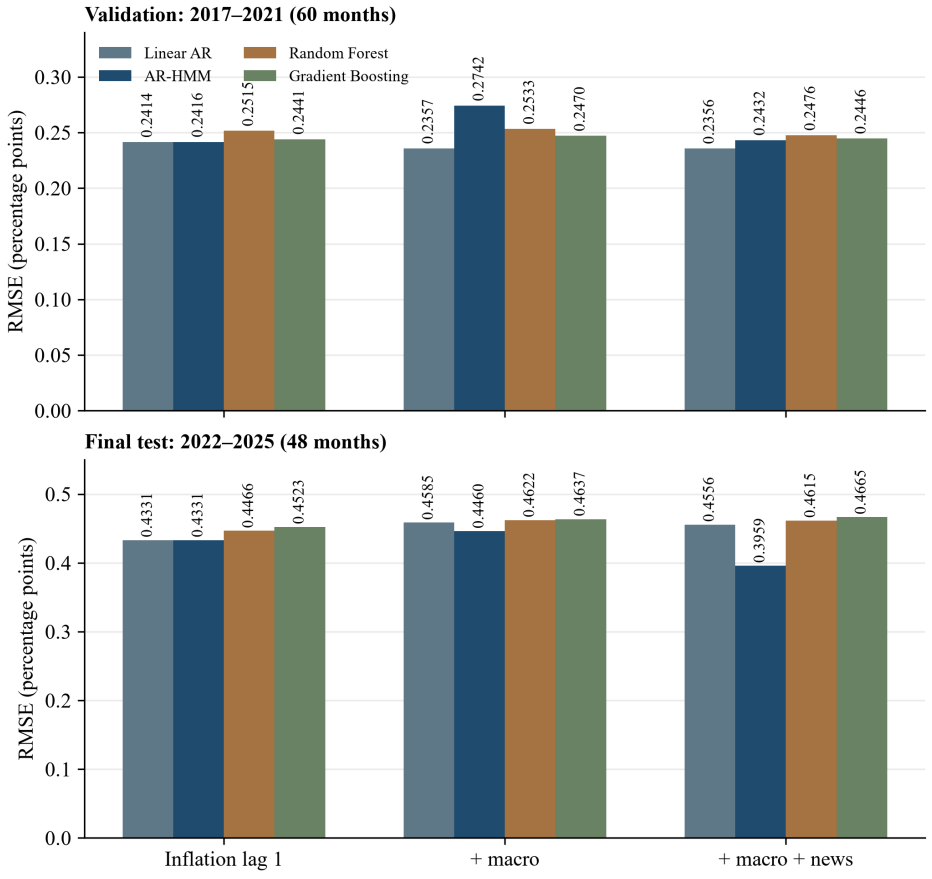



Figure 5.5. Matched European validation and final-test RMSE by information set. Validation: 2017–2021 (60 months); final test: 2022–2025 (48 months). Errors are in percentage points. The full-information AR-HMM has the lowest final-test RMSE, at 0.3959.

The full-information AR-HMM achieves the lowest final-test RMSE among all 12 matched specifications, at 0.3959, with MAE of 0.3059. Adding news to its macroeconomic

specification reduces RMSE from 0.4460 by approximately 11.2%; the gain relative to its inflation-only model is approximately 8.6%. These reductions use archived rounded scores.

The improvement occurs in both 2022 and 2023–2025, with the larger gain in 2022 (Table 5.3). Linear AR ranks first during validation within each information set, whereas the AR-HMM with macroeconomic and news inputs achieves the lowest RMSE over the full test period. The strongest validation result is the linear AR with macroeconomic and news predictors (RMSE 0.2356), rather than the inflation-only AR (0.2414).

One plausible interpretation is that the linear model was sufficient under the generally lower-inflation validation conditions, while state-dependent relationships and the broader information set were more useful during the 2022 surge. With the same macroeconomic and news inputs, AR-HMM RMSE in 2022 is 0.5960, compared with 0.7299 for linear AR. Its advantage persists in 2023–2025, although the difference is smaller (0.3009 versus 0.3149). The results establish these accuracy differences; attributing them specifically to a greater prevalence of a high-inflation latent state would require a separate analysis of state probabilities.

The tree ensembles also struggled during the 2022 inflation surge. Regression trees predict constant values within learned terminal regions and cannot extend a linear trend beyond their learned split thresholds [18]. This creates an extrapolation limitation when inflation moves beyond values represented in the 2007–2021 training sample, as occurred for the calendar-month-centred target during 2022. For Random Forest, averaging leaf predictions bounds the centred forecast by the minimum and maximum centred training targets [18]; the training-based seasonal effect is added back afterwards.

Gradient boosting also combines constant-leaf trees and cannot continue a linear trend outside learned splits, although its additive predictions are not strictly bounded by the original training-target range [18].

Table 5.3a. Fitted-state diagnostics for the full-information European AR-HMM. Fit through December 2021; final-test RMSE 0.3959 percentage points (2022–2025).

| **Parameter / predictor** | **State 1** | **State 2** |
|---|---|---|
| Intercept, c | −0.0431 | 0.0036 |
| Inflation lag 1, φ | 0.4324 | 0.0169 |
| Unemployment change, lag 3 | −0.4141 | 0.0318 |
| Industrial production growth, lag 1 | 0.0233 | −0.0252 |
| Energy HICP inflation, lag 1 | 0.1863 | −0.0430 |
| Energy PPI growth, lag 1 | −0.1726 | 0.1156 |
| Mean news sentiment | −0.0844 | 0.0558 |
| Geopolitical-risk news | −0.1425 | 0.0303 |
| Energy-risk news | 0.1395 | 0.0014 |
| Supply-chain-risk news | 0.2118 | 0.0225 |
| Common innovation variance, $\sigma^2$ | 0.0171 | 0.0171 |

Table 5.3a summarises the fitted-state coefficients. State 1 is short-lived ($p_{11} \approx 0$; expected duration ≈ 1.00 month) and has stronger sensitivity to lagged inflation (+0.4324), energy-risk news (+0.1395) and supply-chain-risk news (+0.2118). State 2 is more persistent ($p_{22}$ = 0.7976; expected duration ≈ 4.94 months) and has little sensitivity to lagged inflation (+0.0169). These are conditional-mean states with common innovation variance, not independently identified crisis or business-cycle regimes.

Mean news sentiment has opposite conditional associations: a one-standard-deviation increase toward more positive coverage corresponds to −0.0844 percentage points in State 1 and +0.0558 in State 2. The negative State 1 estimate could be consistent with adverse coverage accompanying supply-driven inflation pressures, but this mechanism is not identified by the model.

### 5.7 Supplementary U.S. Results

The supplementary U.S. analysis follows the same reporting sequence: inflation baselines, macroeconomic inputs and news additions. Table 5.4 summarises selected results over January 2022–September 2025 (45 months). Unlike Europe, this exercise has no separate validation stage.

Table 5.4. Supplementary U.S. monthly-inflation results, in percentage points. Static fits through 2021. AR-HMM rows distinguish unrestricted switching from variance-only switching.

| Specification | Test RMSE | Test MAE |
|---|---|---|
| Linear AR(1) | 0.263740 | 0.182455 |
| Unrestricted AR-HMM(1) | 0.303487 | 0.197187 |
| Variance-only AR-HMM(1) | 0.261231 | 0.180262 |
| Variance-only + unemployment + industry | 0.261180 | 0.180250 |
| Linear AR + mean sentiment | 0.263938 | 0.183034 |
| Linear AR + sentiment dispersion | 0.261011 | 0.180731 |
| Variance-only + mean sentiment | 0.261701 | 0.181416 |
| Variance-only + all lagged topic risks | 0.274330 | 0.185814 |

The variance-only AR-HMM improves on unrestricted switching and slightly outperforms linear AR. Adding unemployment and industry changes its RMSE only marginally. News additions do not improve the variance-only baseline; linear AR benefits modestly from sentiment dispersion.

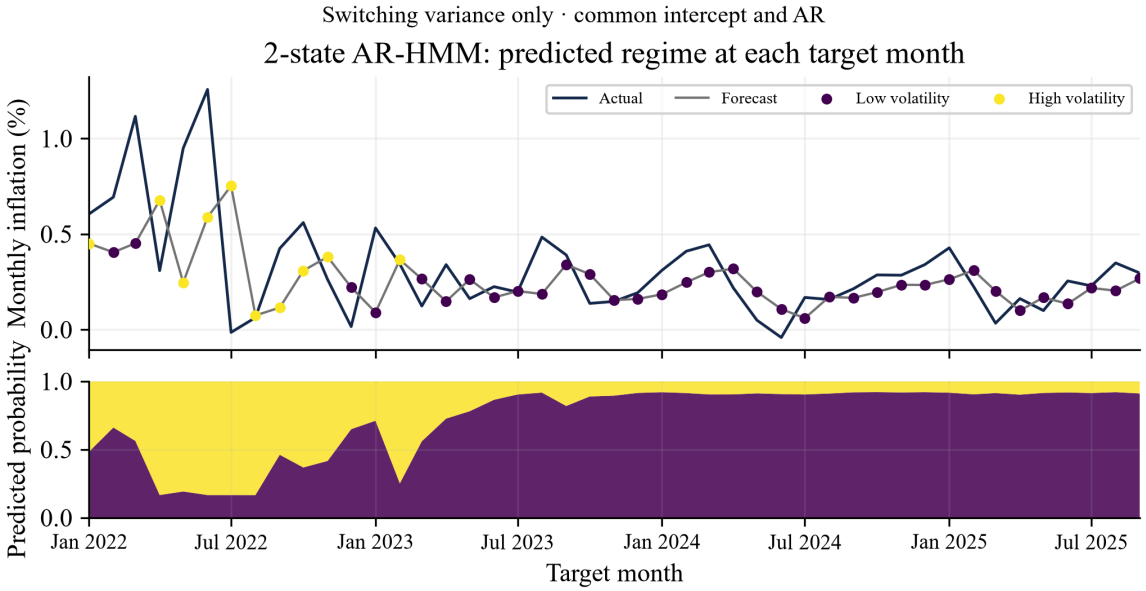



Figure 5.6. U.S. variance-only model estimates, predicted states and probabilities

Figure 5.6 shows higher predicted volatility during much of 2022 and lower volatility later. These variance states differ from the European conditional-mean states.

### 5.8 What the Results Establish

The European analysis demonstrates an incremental predictive contribution from news in the matched AR-HMM: adding the news block to macroeconomic inputs lowers final-test RMSE by approximately 11.2% and produces the lowest error among the 12 matched specifications. The gain is visible in both reported subperiods. This directly answers the thesis's central question about the information added by news.

The tables also show that the benefit emerges from the broader information set: individual sentiment measures and the focused energy combination do not deliver the same improvement. For the company's analytical dashboard, the full-information AR-HMM is therefore a useful specification for further assessment alongside the linear AR benchmark.

### 5.9 Study Limitations

Access to suitable historical news was the main practical constraint. Bloomberg data and API access were unavailable during the thesis, so the analysis relies on Guardian and New York Times articles. Each corpus reflects one English-language publication's selected coverage. The company plans to provide additional resources, allowing future work to assess broader country, sector and event coverage.

Several qualifications apply to interpretation. FinBERT sentiment and dictionary matches have not been validated against a manually annotated inflation-news corpus. The topic indicators combine coverage and negative financial tone; they are not calibrated event probabilities or measures of the direction of inflation pressure. Likewise, the AR-HMM states describe fitted inflation relationships rather than identified structural shocks. The analysis evaluates predictive associations and does not identify causal transmission effects.

The historical data also constrain claims about operational performance. Economic observations are used as downloaded, without reconstructing the releases and vintages available at each past month-end. Calendar lags therefore describe alignment rather than verified publication availability. Testing the dashboard with time-stamped news and economic releases is the next step toward measuring live performance. The U.S. exercise remains supplementary because its data and evaluation design differ from Europe's.

### 5.10 Dashboard Prototype and User Workflow

Global Macro Signals (GMS), the analytical dashboard, brings historical observations, precomputed model estimates and the news corpus into a single interface. The Streamlit prototype reads prepared CSV and JSON files, without a database connection or model re-estimation during use. It supports inspection of the thesis results and discussion with company analysts.

The user first selects Europe or US and a period: the last five years, the full history or a custom interval. Four overview cards show the latest available monthly inflation, annual inflation, unemployment and news sentiment within that interval. They are not period averages; the sentiment value averages article scores within the displayed month. In Historical situations, users switch between annual and monthly inflation and inspect unemployment, industrial production and monthly FinBERT net sentiment. Figure 5.7 illustrates the European overview and annual inflation chart.

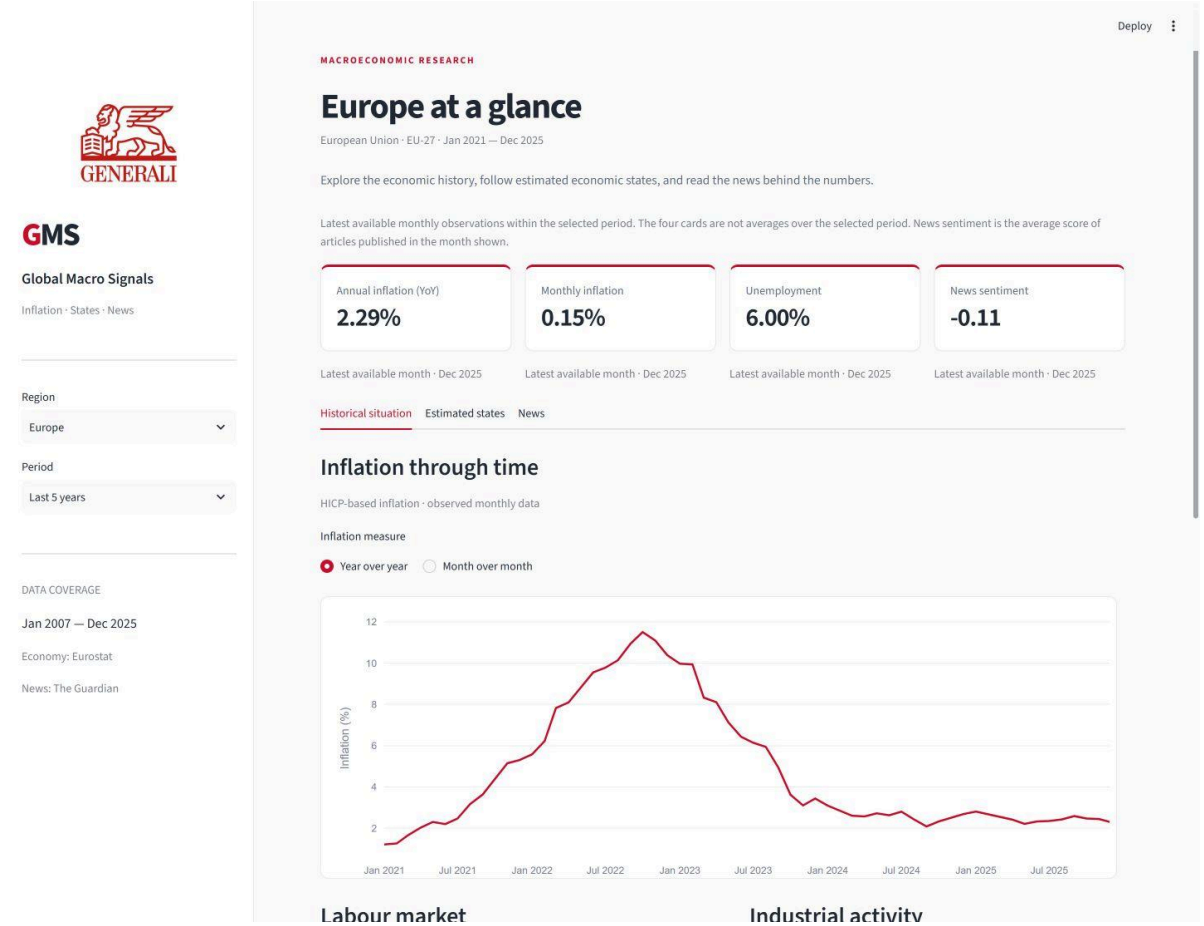



Figure 5.7. GMS overview and Historical situation view for Europe, January 2021–December 2025.

Estimated states connect the regime analysis to observed inflation (Figure 5.8). It displays the latest estimated state, its probability, the monthly inflation estimate, the probability paths and a comparison of observed and estimated inflation. The European view reproduces the calendar-adjusted two-state AR-HMM(1) shown in Figure 4.2, with estimates from January 2022 to December 2025. Its low- and high-inflation labels are relative model regimes, rather than fixed thresholds.

The U.S. view uses the variance-only two-state AR-HMM(1) in Figure 5.6, with estimates through September 2025. Its states denote low and high shock volatility, with a common intercept and autoregressive coefficient. The two regions therefore require different interpretations. In both views, state probabilities are formed before observing the target month's inflation; the displayed probability is not a forecast interval for inflation.

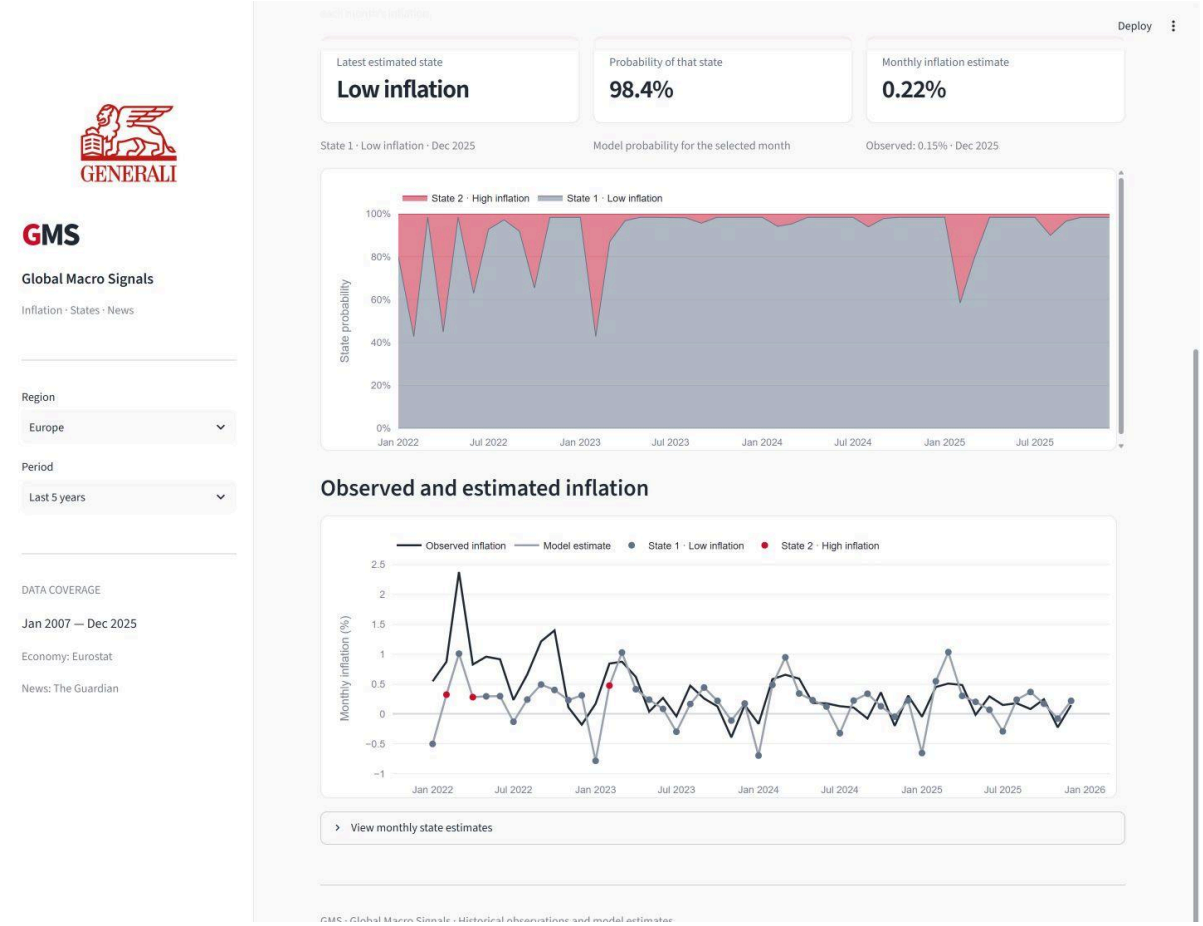



Figure 5.8. European Estimated states view, showing estimated state probabilities and observed versus estimated monthly inflation over 2022–2025.

The News view allows users to investigate the articles behind the aggregate indicators. It presents dated Guardian headlines for Europe and New York Times headlines for the US, with their sentiment labels. Users can search headlines, introductory text or topics, filter by sentiment and sort by publication date. Figure 5.9 shows a search for "inflation" within the selected European period. Selecting a headline opens an article reader containing the available introductory text and, where available, a link to the publisher's complete article. The dashboard does not generate or retrieve new article content.

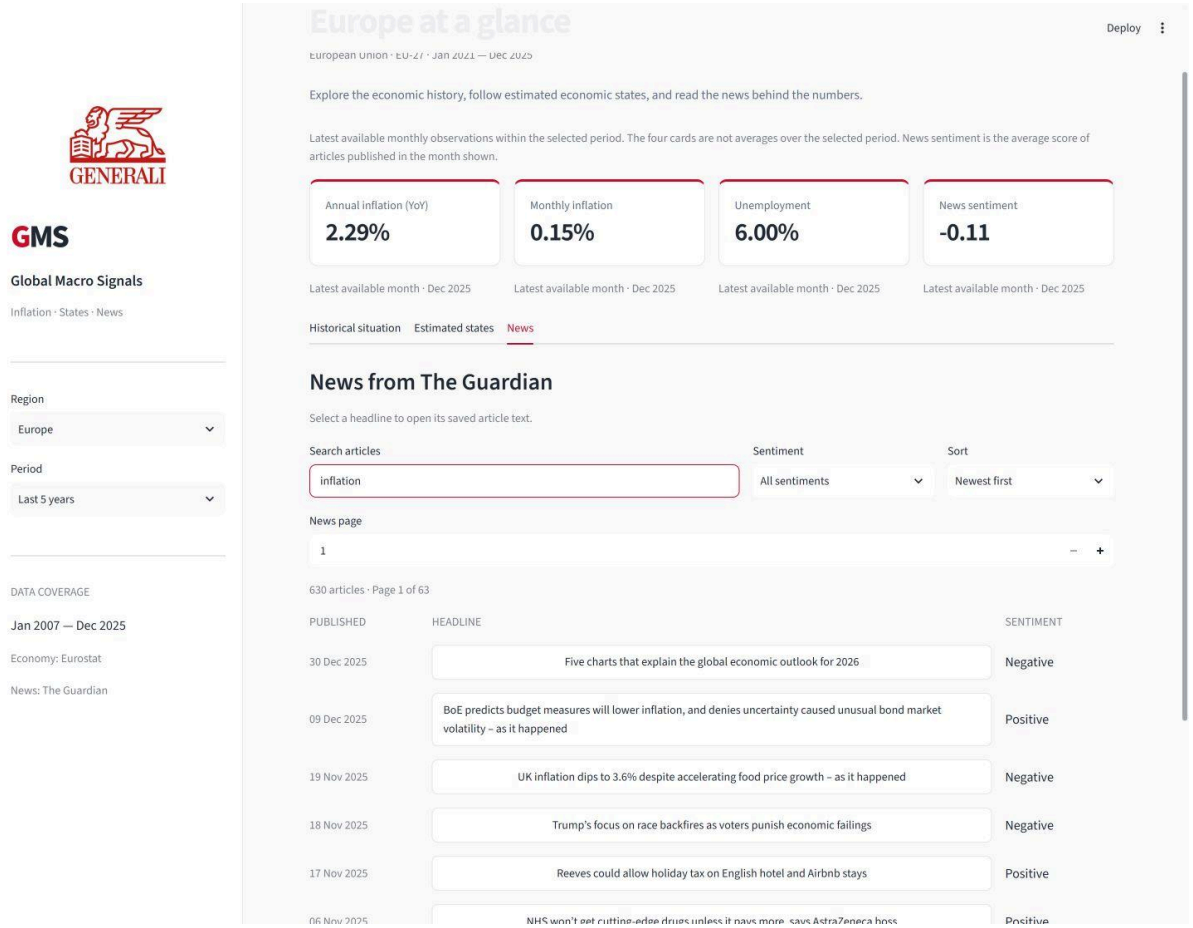



Figure 5.9. European News view with the search term "inflation", sentiment and sorting controls, and dated headline results.

Historical observations and state estimates can be inspected in expandable tables and downloaded as CSV files; the U.S. workbook is also downloadable. Missing observations remain gaps. These interactions demonstrate an implemented research prototype using archived data and estimates. Live ingestion, scheduled estimation at a defined information cut-off and assessment against subsequent releases remain future extensions, as discussed in Section 6.2.

# Chapter 6 Conclusion

## 6.1 Answers to the Research Questions and Contributions

News adds a broader information set to monthly inflation estimation. This thesis tested its contribution alongside regime switching and macroeconomic predictors for EU-27 month-on-month inflation. The principal comparisons combine target-month articles with lagged economic inputs, yielding retrospective month-end nowcasts.

For RQ1, AR-HMM improves accuracy relative to the other model families in the matched final test when it receives the full macroeconomic and news information set. Its 2022–2025 RMSE is 0.3959 percentage points, compared with 0.4556 for linear AR, 0.4615 for Random Forest and 0.4665 for gradient boosting. It has the lowest RMSE among all 12 matched specifications and MAE of 0.3059.

For RQ2, the initial European validation selects two states and one inflation lag under the stated RMSE criterion. More states or longer lag orders do not consistently improve accuracy. Calendar-month centring is useful in the archived seasonal experiment: final-test AR-HMM RMSE falls from 0.4658 without centring to 0.4553 with centring. That experiment retains its own baseline because preprocessing and separately estimated fits differ from the later matched comparison; its scores should not be combined with the matched 0.3959 result to attribute a single cumulative gain.

For RQ3, macroeconomic predictors alone do not consistently improve accuracy beyond inflation lags, but adding the news block provides a clear observed gain within the matched AR-HMM. Its final-test RMSE rises from 0.4331 with inflation alone to 0.4460 with macroeconomic inputs, then falls to 0.3959 when target-month mean sentiment and the three topic-risk indicators are added. The news addition reduces RMSE by 0.0501 percentage points, approximately 11.2%, relative to the macro-only specification. The full-information model is also approximately 8.6% below the inflation-only AR-HMM.

This answer remains specific to the combined information set. The narrower experiments give mixed results: energy variables improve validation but worsen final-test accuracy, unemployment offers a modest gain, generic sentiment does not consistently reduce full-test RMSE, and the focused energy-PPI and energy-risk combination performs worse than its baseline (Sections 5.4–5.5). The full-information AR-HMM improves on its inflation-only version in both 2022 and 2023–2025, with a larger gain in 2022.

The thesis contributes a documented monthly dataset linking EU-27 inflation, macroeconomic series and dated Guardian news; an evaluation framework for comparing model structures and predictor groups on matched dates; and a prototype analytical dashboard for company use. The supplementary U.S. analysis provides exploratory context, with limited and specification-dependent gains from sentiment additions. The main

European finding is that the broader news block improves the matched macroeconomic AR-HMM over the final-test period. This supports retaining that specification for further assessment alongside simple AR benchmarks, with conclusions bounded by the available news sources and retrospective design.

### 6.2 Future Research

The first priority is to use the news resources that the company plans to provide. Their suitability should be checked for historical coverage, usable article content, publication timestamps and consistent access. Once suitable resources are available, they can be connected to the existing preparation and sentiment-scoring workflow. This directly addresses the principal data-access constraint encountered during the thesis.

The next step is to test whether the company-provided sources add predictive value. The initial comparison should retain the current sentiment and topic-risk definitions, model specifications and evaluation periods so that the effect of changing the news source can be assessed clearly. Linear AR and the current full-information AR-HMM should remain reference models. More detailed country or sector filters should be introduced only where the supplied metadata and article coverage support them; broader access should not be assumed to improve accuracy.

A practical operational extension is to connect the supplied news feed to the prototype dashboard and generate estimates at a fixed monthly cut-off. Incoming articles and economic releases should be checked for missing or duplicate records, and each estimate should retain a snapshot of the inputs available at that time. Subsequent comparison with released inflation would then assess performance under actual information availability and show whether the retrospective gains persist in company use.

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# Appendix A  Extended Results and Diagnostics

## A.1 Reading the extended results

European validation covers 2017–2021 (60 months) and testing 2022–2025 (48 months), using calendar-centred inflation. The exploratory U.S. test covers January 2022–September 2025 (45 months), without independent validation. U.S. models fit through 2021 unless labelled expanding.

RMSE, MAE and ME are expressed in percentage points, with ME defined as actual minus estimated inflation. Table A.1 reports four decimal places; the remaining tables report six. Each experiment is interpreted against its own baseline.

### A.2 European extended results

**Table A.1. Matched European model families: validation and final-test accuracy**

| Inputs / family | Val. RMSE | Val. MAE | Test RMSE | Test MAE |
|---|---|---|---|---|
| I / Linear AR | 0.2414 | 0.1613 | 0.4331 | 0.3230 |
| I / AR-HMM | 0.2416 | 0.1610 | 0.4331 | 0.3263 |
| I / Random Forest | 0.2515 | 0.1781 | 0.4466 | 0.3322 |
| I / Gradient Boosting | 0.2441 | 0.1658 | 0.4523 | 0.3400 |
| I+M / Linear AR | 0.2357 | 0.1569 | 0.4585 | 0.3529 |
| I+M / AR-HMM | 0.2742 | 0.1930 | 0.4460 | 0.3479 |
| I+M / Random Forest | 0.2533 | 0.1747 | 0.4622 | 0.3503 |
| I+M / Gradient Boosting | 0.2470 | 0.1677 | 0.4637 | 0.3541 |
| I+M+N / Linear AR | 0.2356 | 0.1532 | 0.4556 | 0.3402 |
| I+M+N / AR-HMM | 0.2432 | 0.1727 | 0.3959 | 0.3059 |
| I+M+N / Random Forest | 0.2476 | 0.1655 | 0.4615 | 0.3467 |
| I+M+N / Gradient Boosting | 0.2446 | 0.1637 | 0.4665 | 0.3440 |

I denote inflation lag 1; M denotes the four macroeconomic inputs at the lags in Table 4.2; N denotes target-month mean sentiment and three topic indicators. MAE complements the RMSE results in Table 5.3.

**Table A.2a. European fixed-lag macroeconomic inputs**

| Added inputs | Val. RMSE | Test RMSE | Test MAE | Test ME |
|---|---|---|---|---|
| Baseline | 0.241558 | 0.433122 | 0.326277 | 0.156324 |
| + unemployment | 0.241450 | 0.430356 | 0.320507 | 0.160460 |
| + industrial production | 0.243411 | 0.434226 | 0.326379 | 0.156287 |
| + energy HICP | 0.239292 | 0.467408 | 0.339938 | 0.206688 |
| + energy PPI | 0.226721 | 0.459437 | 0.356988 | 0.230968 |
| + energy block | 0.224659 | 0.453766 | 0.351012 | 0.224491 |
| + slow macro | 0.250581 | 0.435583 | 0.325532 | 0.162293 |
| + all macro | 0.233467 | 0.450851 | 0.352108 | 0.223627 |

All macroeconomic lags are 1. Energy block: energy HICP and PPI; slow macro: unemployment and industrial production. RMSE summary: Table 5.1.

**Table A.2b. European target-month news indicators**

| Added inputs | Val. RMSE | Test RMSE | Test MAE | Test ME |
|---|---|---|---|---|
| Baseline | 0.240849 | 0.433122 | 0.326277 | 0.156324 |
| + mean sentiment | 0.240487 | 0.437417 | 0.331426 | 0.159011 |
| + median sentiment | 0.243358 | 0.434684 | 0.329155 | 0.154899 |
| + negative share | 0.246152 | 0.433505 | 0.330174 | 0.152270 |
| + sentiment dispersion | 0.241725 | 0.438252 | 0.324025 | 0.167199 |
| + geopolitical risk | 0.248537 | 0.447707 | 0.333311 | 0.187185 |
| + energy risk | 0.239496 | 0.433661 | 0.328784 | 0.142512 |
| + supply-chain risk | 0.240711 | 0.426314 | 0.313789 | 0.149190 |
| + all subject risks | 0.255729 | 0.415643 | 0.307939 | 0.122422 |

RMSE summary: Table 5.2.

**Table A.3. European validation-selected macro lags**

| Specification | Validation RMSE | Test RMSE | Test MAE | Test ME |
|---|---|---|---|---|
| Baseline | — | 0.433122 | 0.326277 | 0.156324 |
| + unemployment lag 3 | 0.236519 | 0.427897 | 0.318896 | 0.150488 |
| + industrial production lag 1 | 0.242578 | 0.442905 | 0.344964 | 0.156083 |
| + energy HICP lag 1 | 0.234209 | 0.467399 | 0.339898 | 0.206727 |
| + energy PPI lag 1 | 0.226721 | 0.459437 | 0.356989 | 0.230968 |
| + selected macro lags | 0.244375 | 0.464662 | 0.359972 | 0.232087 |

Lags are selected from 1–6, and the joint model combines all four macroeconomic indicators. The baseline serves as a final-test reference; its validation RMSE is not reported (—). This lag-selection experiment is separate from the matched comparison in Table A.1.

**Table A.4. Focused European energy-PPI and energy-risk comparison**

| Specification | Validation RMSE | Test RMSE | Test MAE | Test ME |
|---|---|---|---|---|
| Baseline | 0.240849 | 0.433122 | 0.326277 | 0.156324 |
| + energy PPI lag 1 | 0.226721 | 0.459438 | 0.356989 | 0.230969 |
| + energy-risk sentiment | 0.239496 | 0.433662 | 0.328784 | 0.142514 |
| + macro and sentiment | 0.266516 | 0.474285 | 0.358637 | 0.225282 |

Energy PPI uses lag 1; energy risk uses the target month. This focused design differs from the matched nine-predictor comparison (Section 5.6).

## A.3 Supplementary U.S. results

**Table A.5. U.S. baseline, expanding estimation and switching restrictions**

| Specification | RMSE | MAE | ME |
|---|---|---|---|
| Static AR(1) | 0.263740 | 0.182455 | 0.067387 |
| Static ARX + sentiment | 0.263938 | 0.183034 | 0.069157 |
| Static HMM | 0.302536 | 0.197173 | 0.142413 |
| Static AR-HMM | 0.303487 | 0.197187 | 0.053888 |
| Static AR-HMM + lagged sentiment | 0.261486 | 0.181474 | 0.070805 |
| Static LASSO | 0.267084 | 0.185420 | 0.083598 |
| Static LASSO + sentiment | 0.267930 | 0.183599 | 0.099399 |
| Expanding AR(1) | 0.265544 | 0.181674 | 0.054634 |
| Expanding AR-HMM(1) | 0.267127 | 0.185508 | 0.063486 |
| Switching intercept · common AR · common variance | 0.261659 | 0.180324 | 0.074924 |
| Switching intercept + variance · common AR | 0.264829 | 0.181883 | 0.072158 |
| Switching variance only · common intercept and AR | 0.261231 | 0.180262 | 0.063102 |

Expanding models refit monthly. HMM has two state-specific means, common variance and no inflation lags. LASSO (least absolute shrinkage and selection operator) uses inflation lags 1–6, adding sentiment lags 1–6 in its extension; the penalty is selected by chronological cross-validation within training data. ARX uses target-month sentiment; the AR-HMM sentiment extension uses lag 1. Selected results appear in Table 5.4.

**Table A.6. U.S. monthly lag-order comparison**

| Specification | RMSE | MAE | ME |
|---|---|---|---|
| Static AR(1) | 0.263740 | 0.182455 | 0.067387 |
| MS-Variance AR(1) | 0.261231 | 0.180262 | 0.063102 |
| AR(2) | 0.276363 | 0.192375 | 0.085438 |
| MS-Variance AR(2) | 0.271161 | 0.186523 | 0.089523 |
| AR(3) | 0.276341 | 0.192362 | 0.085417 |
| MS-Variance AR(3) | 0.277484 | 0.189833 | 0.095695 |
| AR(4) | 0.276972 | 0.192142 | 0.079834 |
| MS-Variance AR(4) | 0.270144 | 0.184350 | 0.071245 |
| AR(5) | 0.283592 | 0.195195 | 0.089243 |
| MS-Variance AR(5) | 0.268158 | 0.183366 | 0.065695 |
| AR(6) | 0.284635 | 0.195592 | 0.092447 |
| MS-Variance AR(6) | 0.280838 | 0.190575 | 0.094274 |
| naive | 0.304674 | 0.200200 | -0.008787 |

MS-Variance has common intercept and slopes with state-specific innovation variance. The naïve estimate uses the previous month's inflation.

**Table A.7. U.S. monthly inflation: all 16 macroeconomic information sets**

| Added macro inputs | AR RMSE | MS-var. RMSE | AR MAE | MS-var. MAE |
|---|---|---|---|---|
| Inflation lag 1 only | 0.263740 | 0.261231 | 0.182455 | 0.180262 |
| Oil | 0.277795 | 0.277911 | 0.202608 | 0.200310 |
| Unemployment | 0.263928 | 0.261268 | 0.182532 | 0.180310 |
| Industry | 0.263753 | 0.261216 | 0.182372 | 0.180250 |
| Fed rate | 0.286857 | 0.261561 | 0.208379 | 0.183880 |
| Oil + Unemployment | 0.282465 | 0.282597 | 0.207562 | 0.206024 |
| Oil + Industry | 0.280415 | 0.277069 | 0.205540 | 0.199110 |
| Oil + Fed rate | 0.275599 | 0.277015 | 0.204555 | 0.200238 |
| Unemployment + Industry | 0.264054 | 0.261180 | 0.182832 | 0.180250 |
| Unemployment + Fed rate | 0.297396 | 0.261549 | 0.217248 | 0.183856 |
| Industry + Fed rate | 0.296181 | 0.261571 | 0.215252 | 0.183982 |
| Oil + Unemployment + Industry | 0.282325 | 0.282485 | 0.207525 | 0.205954 |
| Oil + Unemployment + Fed rate | 0.284570 | 0.278313 | 0.212759 | 0.204343 |
| Oil + Industry + Fed rate | 0.283106 | 0.275551 | 0.209407 | 0.200097 |
| Unemployment + Industry + Fed rate | 0.298173 | 0.261489 | 0.217574 | 0.183886 |
| Oil + Unemployment + Industry + Fed rate | 0.285189 | 0.278013 | 0.212340 | 0.204161 |

All subsets include inflation lag 1; macroeconomic inputs also use lag 1. MS-var. denotes the variance-only structure defined below Table A.6.

**Table A.8. U.S. monthly sentiment extensions**

| Specification | RMSE | MAE | ME |
|---|---|---|---|
| Static AR(1) | 0.263740 | 0.182455 | 0.067387 |
| Static ARX + sentiment | 0.263938 | 0.183034 | 0.069157 |
| AR(1) + Industry | 0.263753 | 0.182372 | 0.066789 |
| AR(1) + Industry + sentiment | 0.263969 | 0.183095 | 0.068537 |
| MS-Variance AR(1) | 0.261231 | 0.180262 | 0.063102 |
| MS-Variance AR(1) + sentiment | 0.261701 | 0.181416 | 0.066483 |
| MS-Variance AR(1) + Unemployment + Industry | 0.261180 | 0.180250 | 0.063726 |
| MS-Variance AR(1) + Unemployment + Industry + sentiment | 0.261672 | 0.181449 | 0.067465 |
| AR(1) + median sentiment | 0.262983 | 0.181389 | 0.063065 |
| AR(1) + negative share | 0.263294 | 0.181940 | 0.065019 |
| AR(1) + sentiment dispersion | 0.261011 | 0.180731 | 0.059421 |
| MS-Variance AR(1) + median sentiment | 0.263747 | 0.183754 | 0.074704 |
| MS-Variance AR(1) + negative share | 0.263567 | 0.182946 | 0.075192 |
| MS-Variance AR(1) + sentiment dispersion | 0.262231 | 0.180881 | 0.066905 |

News measures use the target month; industry and unemployment use lag 1. Dispersion is the sample standard deviation of article-level net sentiment.

**Table A.9. U.S. lagged subject-risk experiment**

| Specification | RMSE | MAE | ME |
|---|---|---|---|
| Static AR(1) | 0.263740 | 0.182455 | 0.067387 |
| AR(1) + geopolitical risk | 0.262466 | 0.181354 | 0.064620 |
| AR(1) + energy risk | 0.277336 | 0.186378 | 0.090952 |
| AR(1) + supply-chain risk | 0.265324 | 0.183463 | 0.069372 |
| AR(1) + all subject risks | 0.275076 | 0.184033 | 0.086929 |
| MS-Variance AR(1) | 0.261231 | 0.180262 | 0.063102 |
| MS-Variance AR(1) + geopolitical risk | 0.262384 | 0.181056 | 0.066281 |
| MS-Variance AR(1) + energy risk | 0.269213 | 0.182300 | 0.081396 |
| MS-Variance AR(1) + supply-chain risk | 0.263819 | 0.181574 | 0.070967 |
| MS-Variance AR(1) + all subject risks | 0.274330 | 0.185814 | 0.091666 |
| Static AR-HMM | 0.262183 | 0.181303 | 0.074565 |
| AR-HMM(1) + geopolitical risk | 0.271333 | 0.190991 | 0.091366 |
| AR-HMM(1) + energy risk | 0.278929 | 0.183632 | 0.101043 |
| AR-HMM(1) + supply-chain risk | 0.264284 | 0.183961 | 0.078634 |
| AR-HMM(1) + all subject risks | 0.284345 | 0.197615 | 0.104262 |

Topic indicators enter at lag 1. The unrestricted AR-HMM was estimated in a separate run from the model in Table A.5. Comparisons within this table therefore use the corresponding baseline reported here.